



Hey, where is **Blue Team**?!

How we are defending AI against side-channel attacks

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Lille (France)
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Credits



Halo Blue team helmet.

Master Replicas. Microsoft.

<https://www.entertainmentearth.com/product/halo-3-master-chief-scaled-blue-helmet-replica/mrms78512?srsId=AfmBOooZr604Me9aVWdv-A58zK2YkeAvnTY3HrB29NSEiB9JSsQ2VPB1>



Baby Duck Plushie.

Kurzgesagt.

<https://shop-us.kurzgesagt.org/collections/plushies/products/baby-duck-plushie?variant=32332051021872>



Vector icons.

FlatIcon (too many artists to list ...)

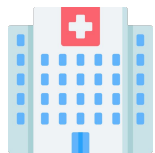
<https://www.flaticon.com/>



Disney Logo.

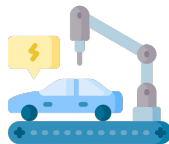
The Walt Disney Company.

Artificial Intelligence: A Pervasive Technology



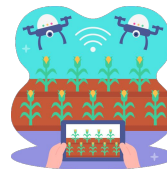
Healthcare [1]

- Disease prediction
- Disease detection
- Precision medicine
- Health monitoring



Automotive [2]

- Autonomous Driving
- Predictive maintenance
- Battery monitoring



Agriculture [3]

- Crop monitoring
- Disease detection
- Yield prediction
- Irrigation tuning



Cybersecurity

- Trojan detection [4]
- Vulnerability identification [5, 6]
- Exploit creation [5, 6]

[1] Deep learning for healthcare: review, opportunities and challenges. R. Miotto, F. Wang, S. Wang, X. Jiang and J. T. Dudley.

[2] The Use of Machine Learning in the Automotive Industry. I3CEET 2024.

[3] A review of deep learning techniques used in agriculture. I. Attri, L. K. Awasthi, T. P.I Sharma, P. Rathee.

[4] Hardware Trojan Detection Using Machine Learning: A Tutorial. <https://dl.acm.org/doi/full/10.1145/3579823>

[5] Assessing Claude Mythos Preview's cybersecurity capabilities. <https://red.anthropic.com/2026/mythos-preview/>

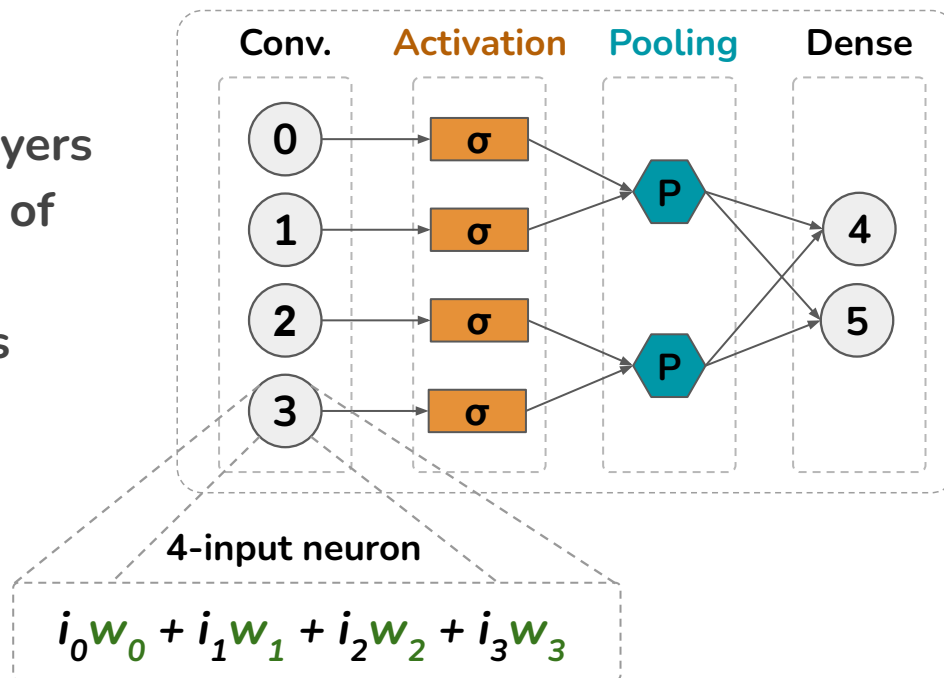
[6] GREYVIBE: A Russia-nexus group leveraging AI across state-aligned operations. <https://labs.withsecure.com/publications/greyvibe>

Deep Neural Networks (DNNs)

Hyperparameters

- # Layers
- Type of layers
- Sequence of layers
- # Neurons

4-layer Convolutional NN (CNN)



Parameters

- Neuron's weights

Deep Neural Networks (DNNs)

Hyperparameters

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Parameters

- Neuron's weights

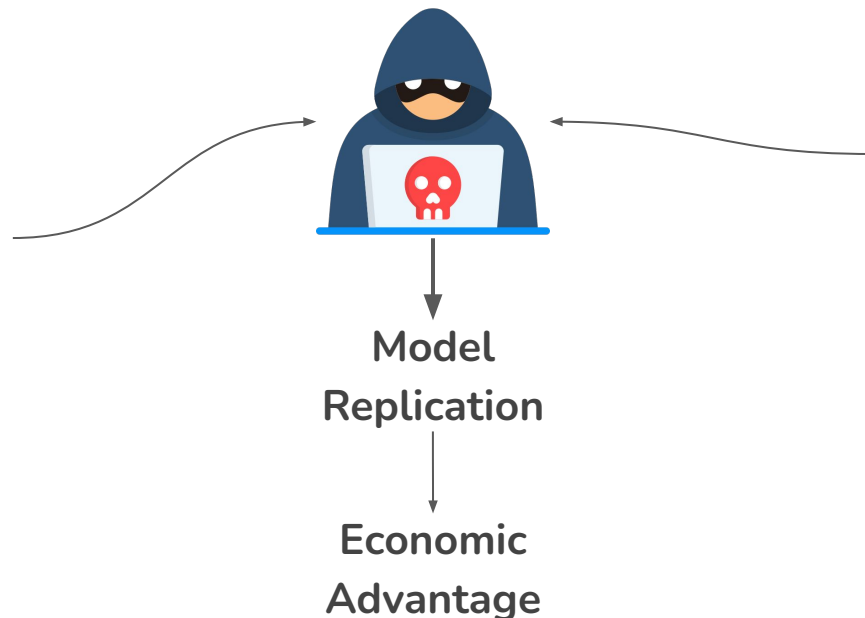
Deep Neural Networks (DNNs)

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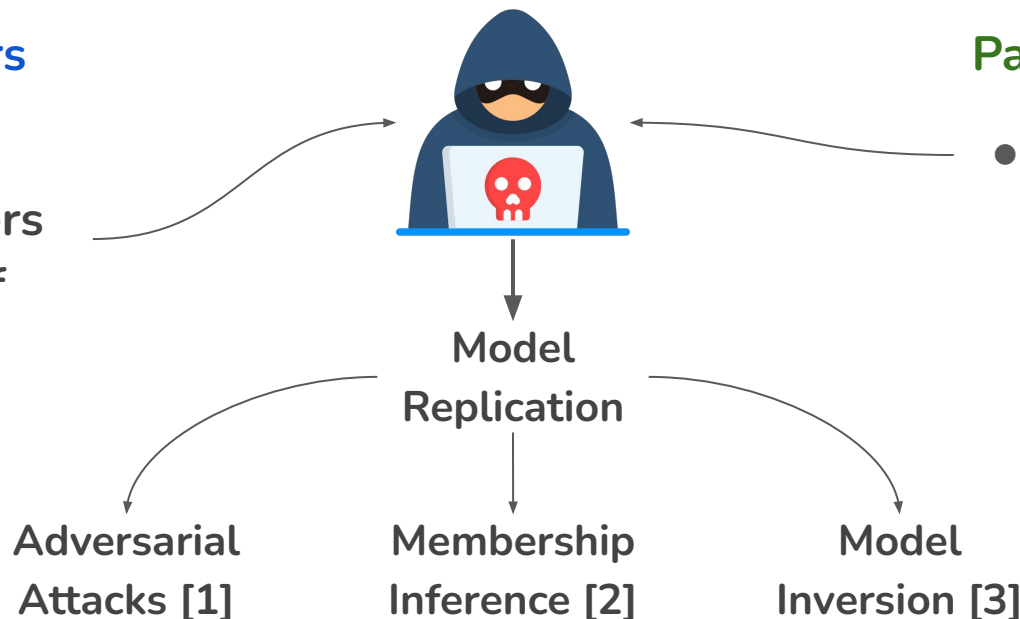
Deep Neural Networks (DNNs)

Hyperparameters

- # Layers
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Parameters

- Neuron's weights



[1] *Intriguing properties of neural networks*. Szegedy C. et al. ICLR 2014.

[2] *Membership Inference Attacks Against Machine Learning Models*. Shokri R. et al. IEEE S&P 2017.

[3] *Model Inversion Attacks that Exploit Confidence Information and Basic Countermeasures*. Fredrikson M. et al. ACM CCS 2015.

**Confidentiality
of (hyper)parameters
is of utmost importance**

but ...

AI at the edge ... and the side-channel threat

Edge Devices

Smartphone



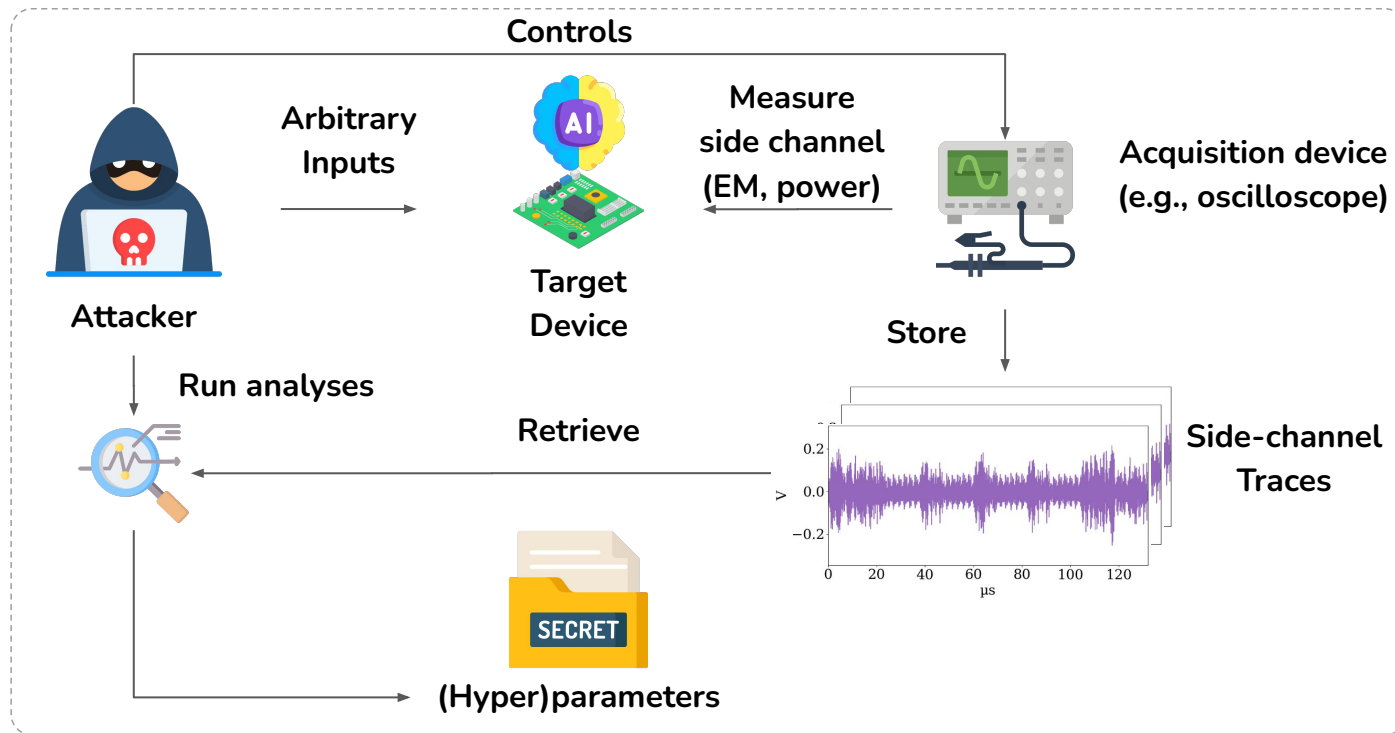
Smartwatch



uControllers



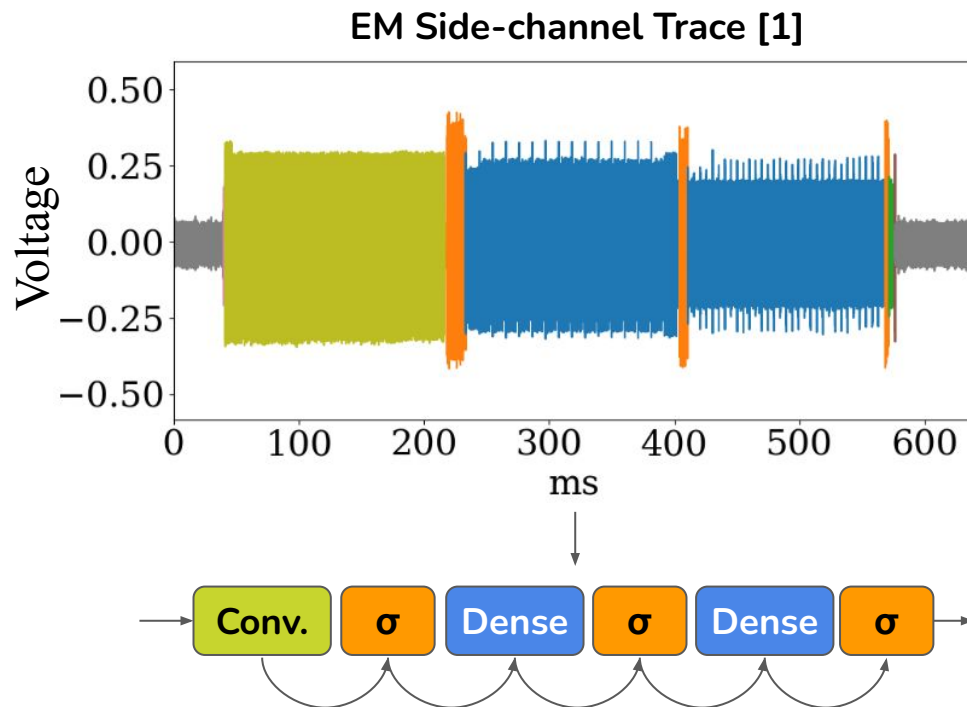
Side-channel Analysis – Threat Model



(Hyper)parameters Recovery

Hyperparameters recovery

- Hyperparameters *influence* side-channel shape, e.g.:
 - Layer types and sequences
 - Number of neurons
 - Convolution features information
- Recovery approach:
 - Simple power analysis

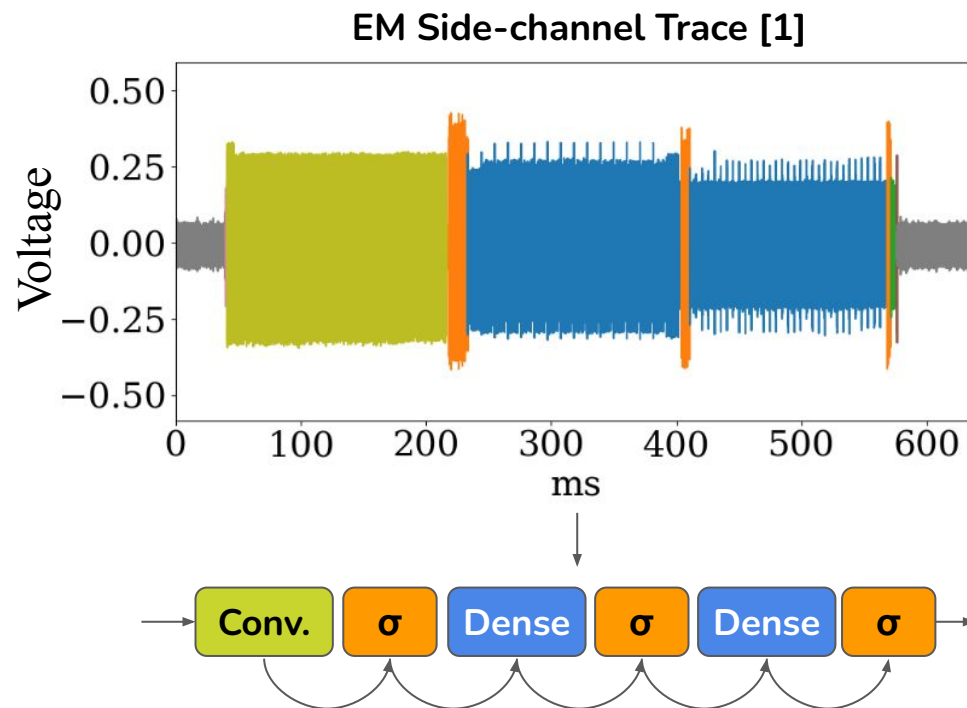


[1] Like an Open Book? Read Neural Network Architecture with Simple Power Analysis on 32-bit Microcontrollers. Joud R. et al. CARDIS 2023

[2] TPUXtract: An Exhaustive Hyperparameter Extraction Framework. Kurian A. et al. IACR CHES 2024

Hyperparameters recovery

- Hyperparameters *influence* side-channel shape, e.g.:
 - Layer types and sequences
 - Number of neurons
 - Convolution features information
- Recovery approach:
 - Simple power analysis
- Recovery demonstrated on:
 - Microcontrollers [1]
 - Tensor product units [2]
- Attack extension:
 - State-of-the-art DNN models [2]
 - Hundreds of layers [2]

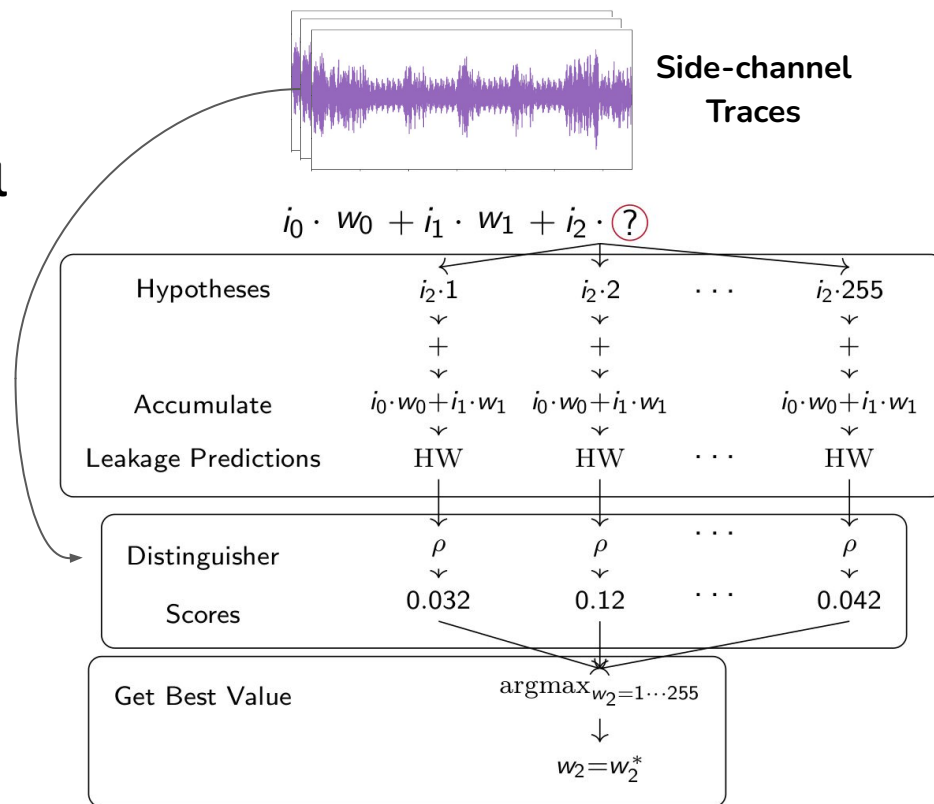


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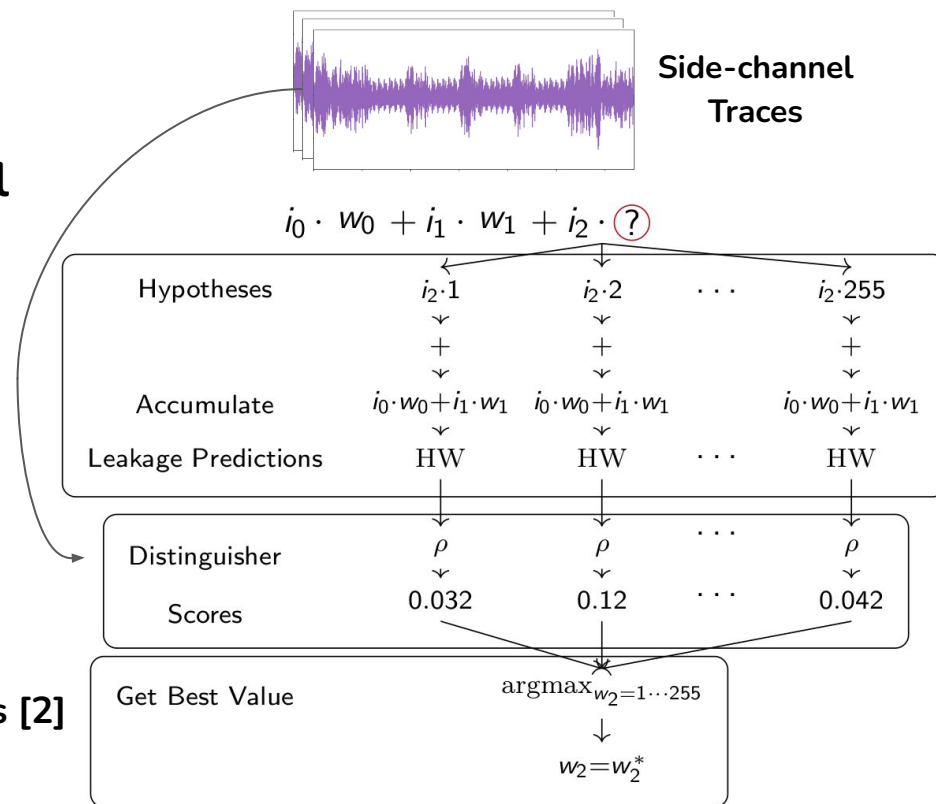
Parameters recovery

- Parameters *influence* side-channel distribution
- Recovery approach:
 - Differential power analysis



Parameters recovery

- Parameters *influence* side-channel distribution
- Recovery approach:
 - Differential power analysis
- Recovery demonstrated on:
 - Microcontrollers [1]
 - GPU units [2, 3]
 - Hardware accelerators [4]
- Attack extension
 - State-of-the-art CNN's first two layers [2]
 - Ten thousands of parameters [2]
 - LLM (proof of concept) [3]



[1] CSI-NN: Reverse Engineering of Neural Network Architectures Through Electromagnetic Side Channel. Batina et al. USENIX 2019.

[2] BarraCUDA: Edge GPUs do Leak DNN Weights. Horvath et al. USENIX 2025.

[3] Kraken: Higher-order EM Side-Channel Attacks on DNNs in Near and Far Field. Horvath et al. Preprint.

Side-channel Countermeasures for Deep Neural Networks

Countermeasure Approaches

Countermeasure Approaches

Classic Approaches

- Generic
- Well studied

Classic Approaches

Masking
(parameters)



Shuffling
(parameters)



Countermeasure Approaches

Classic Approaches

- Generic
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Masking
(parameters)



Shuffling
(parameters)

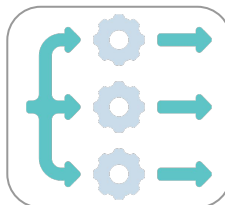


Recent Approaches

Recent Approaches

- DNN-tailored
- First steps

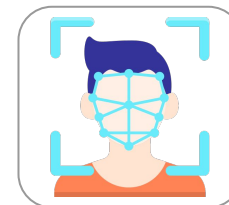
Parallelism
(parameters)



Approximate
Computing
(parameters)



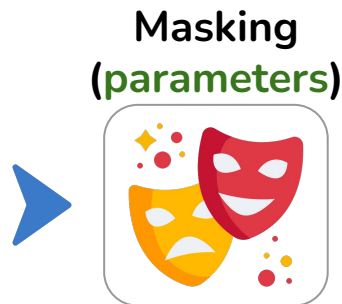
Adversarial Attacks
(hyperparameters)



Countermeasure Approaches

Classic Approaches

- Generic
- Well studied

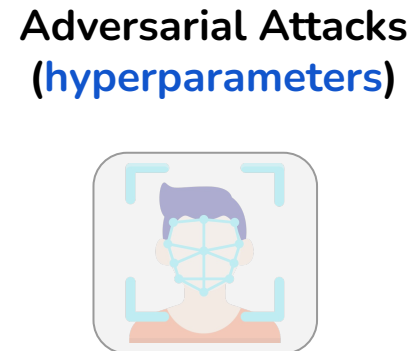
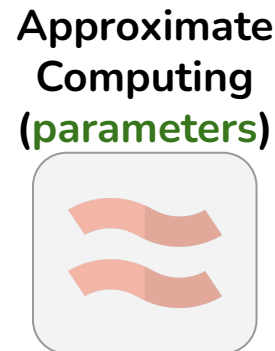
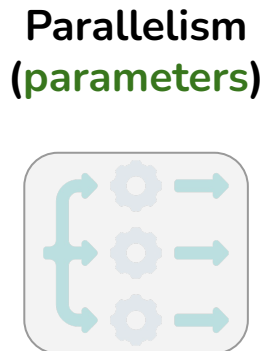


Classic Approaches

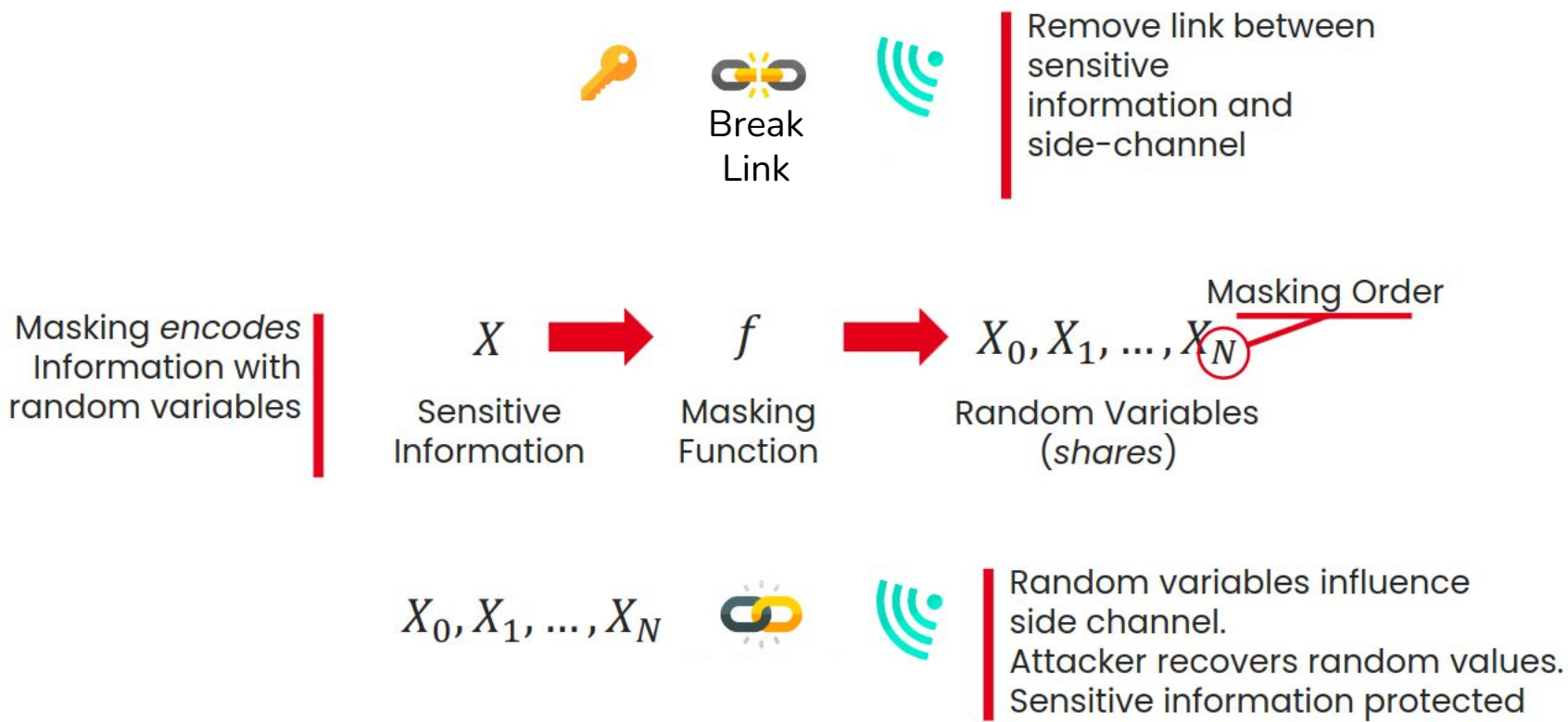
Recent Approaches

Recent Approaches

- DNN-tailored
- First steps



Masking – Principle

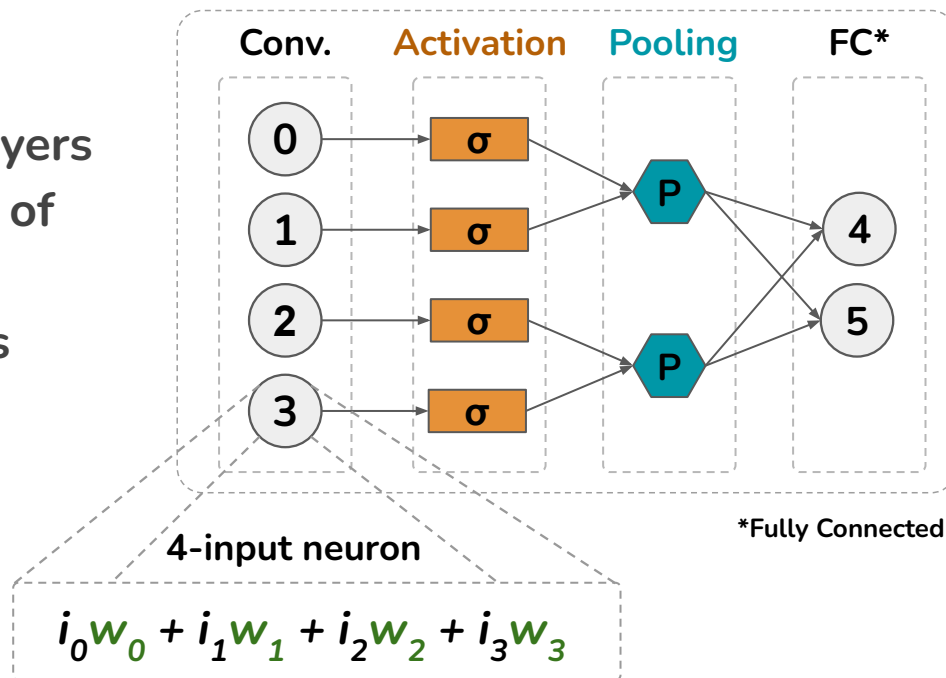


Deep Neural Networks (DNNs)

Hyperparameters

- # Layers
- Type of layers
- Sequence of layers
- # Neurons

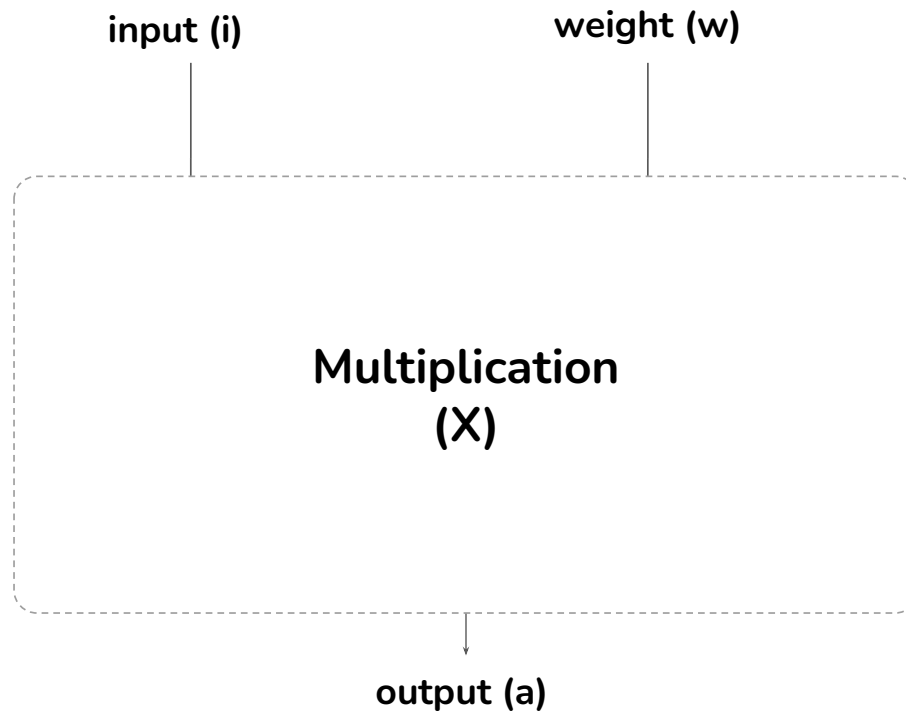
4-layer Convolutional NN (CNN)



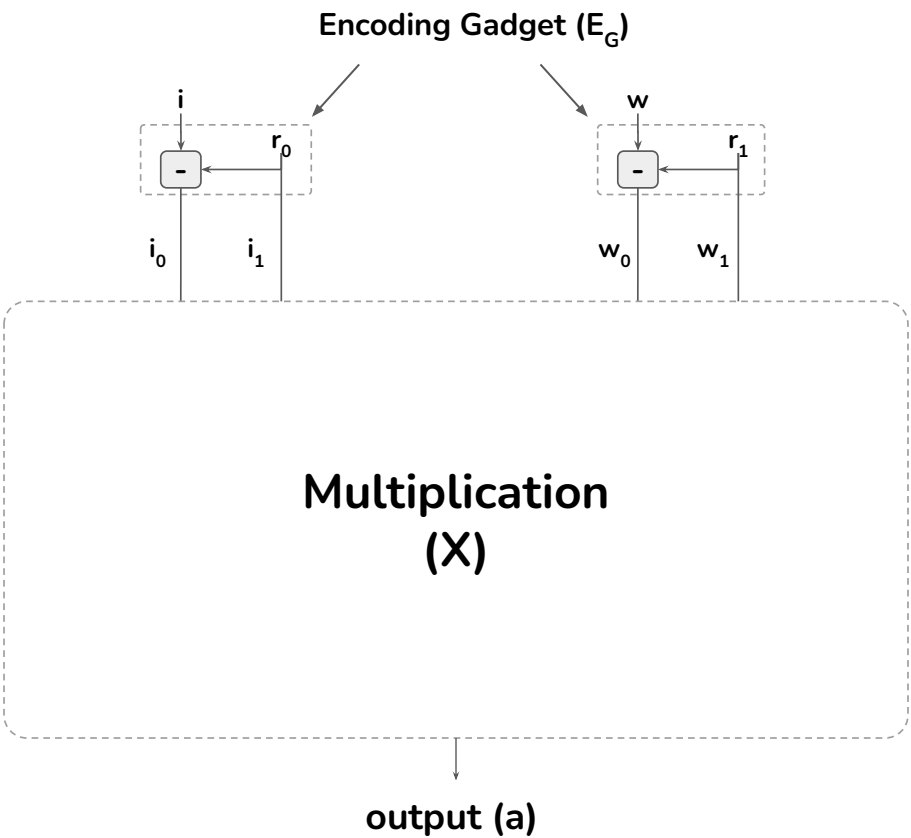
Parameters

- Neuron's weights

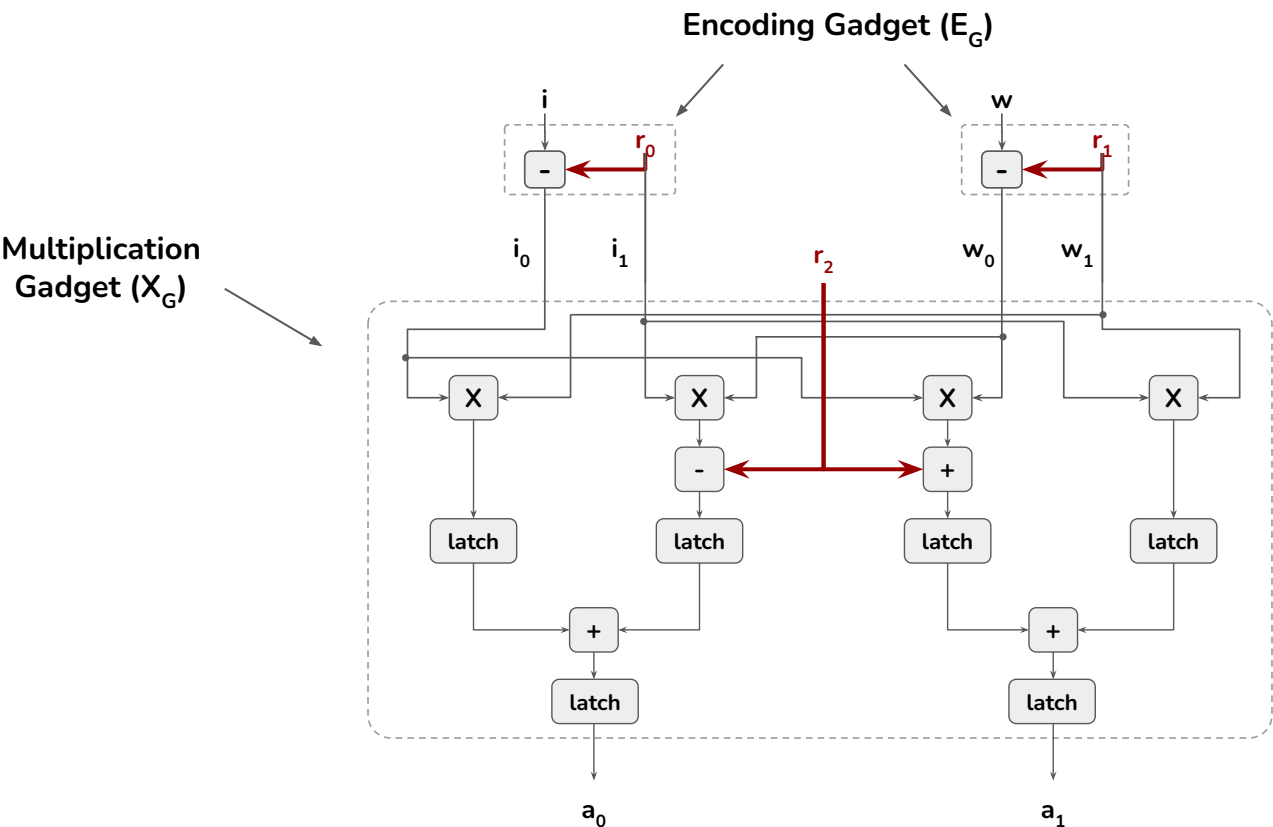
Masking – An Example



Masking – An Example

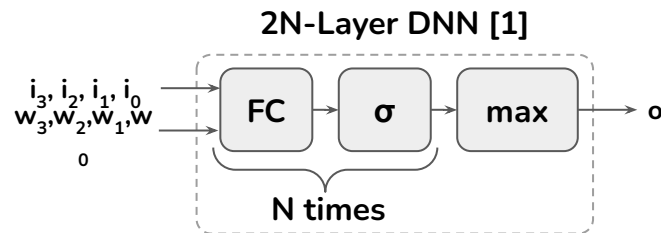


Masking – An Example [1]



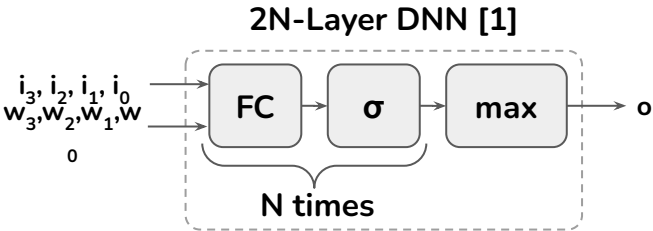
[1] A Masked Hardware Accelerator for Feed-Forward Neural Networks With Fixed-Point Arithmetic. Brosch et al. IEEE VLSI 2023.

Masking – Modular Approach to Security



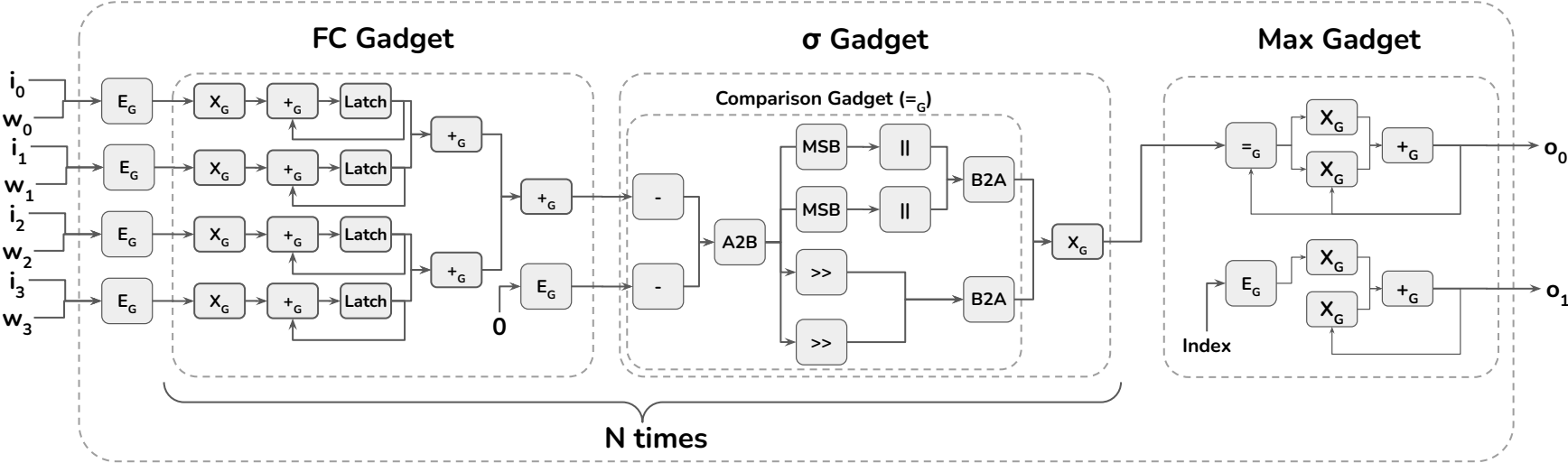
[1] A Masked Hardware Accelerator for Feed-Forward Neural Networks With Fixed-Point Arithmetic. Brosch et al. IEEE VLSI 2023.

Masking – Modular Approach to Security



[1] A Masked Hardware Accelerator for Feed-Forward Neural Networks With Fixed-Point Arithmetic. Brosch et al. IEEE VLSI 2023.

2N-Layer DNN – 1st-order Masked Version [1]



Masking – Advantages and Disadvantages

Advantages:



Provable security

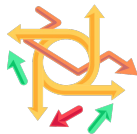
Masking – Advantages and Disadvantages

Advantages:



Provable security

Disadvantages:



Complicate design

New data types	New operations
Floating point Fixed point Signed 1/8/16/32 bits	Comparison Truncation Average

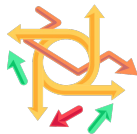
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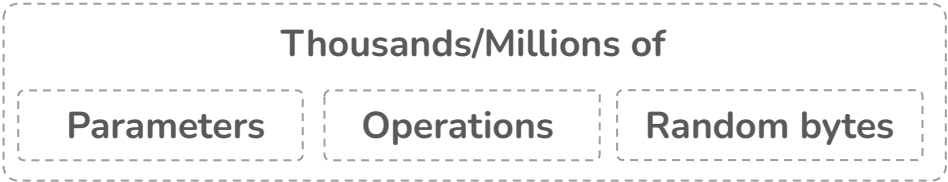
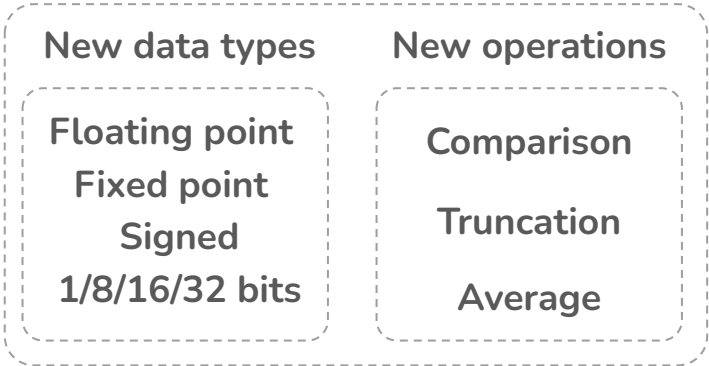
Disadvantages:



Complicate design

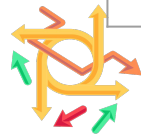


Expensive Design



Masking – Advantages and Disadvantages

Work	Bits	LUTs ($\times 10^3$)	FFs ($\times 10^3$)	Latency ($\times 10^3$ Cycles)	Rand (MiB)
[1]	1	1.83	1.13	2,850.0	0
[1]	1	+437.16%	+574.69%	+3.16 %	25.57



Complicate design

New data types

Floating point
Fixed point
Signed
1/8/16/32 bits

New operations

Comparison

Truncation

Average



Expensive Design

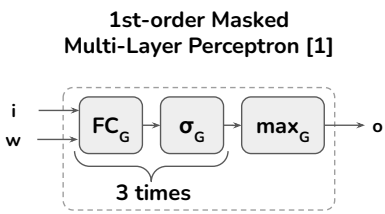
Thousands/Millions of

Parameters

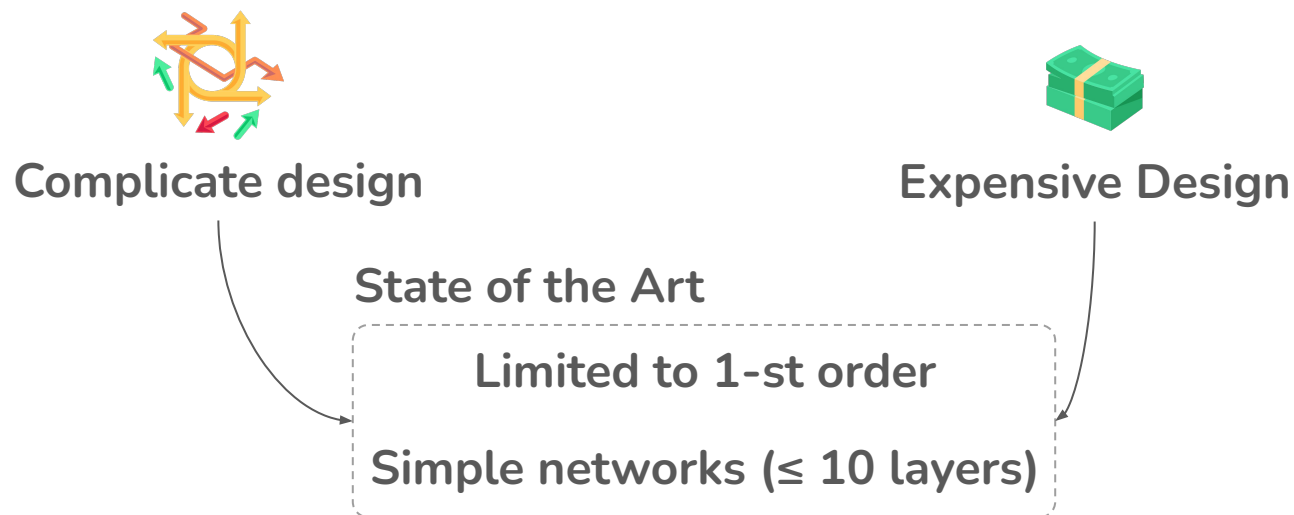
Operations

Random bytes

[1] BoMaNet: Boolean Masking of an Entire Neural Network. Aysu et al. IEEE/ACM ICCAD 2020.



Masking – Research Leads



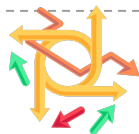
Masking – Research Leads

Alter algebraic structure of DNNs [1]

Standard arithmetic $\rightarrow$ modulo arithmetic

Masking-friendly DNNs

Readapt all the layers and training methodologies



Complicate design



Expensive Design

State of the Art

Limited to 1-st order

Simple networks (≤ 10 layers)

[1] ModuloNET: Neural Networks Meet Modular Arithmetic for Efficient Hardware Masking. Dubey et al. IACR TCHES 2024.

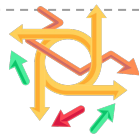
Masking – Research Leads

Alter algebraic structure of DNNs [1]

Standard arithmetic -> modulo arithmetic

Masking-friendly DNNs

Readapt all the layers and training methodologies



Complicate design

Partial masking (Work in progress)

Greatly reduce the overhead [2]

Potentially exposed to other classes of attacks [3]



Expensive Design

State of the Art

Limited to 1-st order

Simple networks (≤ 10 layers)

[1] ModuloNET: Neural Networks Meet Modular Arithmetic for Efficient Hardware Masking. Dubey et al. IACR TCHES 2024.

[2] MaskedNet: The First Hardware Inference Engine Aiming Power Side-Channel Protection. Dubey et al. IEEE HOST 2020.

[3] SITM: See-In-The-Middle Side-Channel Assisted Middle Round Differential Cryptanalysis on SPN Block Ciphers. Bhasin S. et al. IACR TCHES 2020

Countermeasure Approaches

Classic Approaches

- Generic
- Well studied

Classic Approaches

Masking
(parameters)



Shuffling
(parameters)

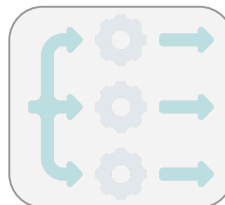


Recent Approaches

Recent Approaches

- DNN-tailored
- First steps

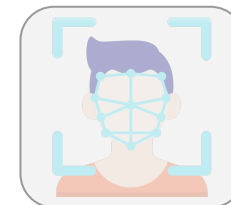
Parallelism
(parameters)



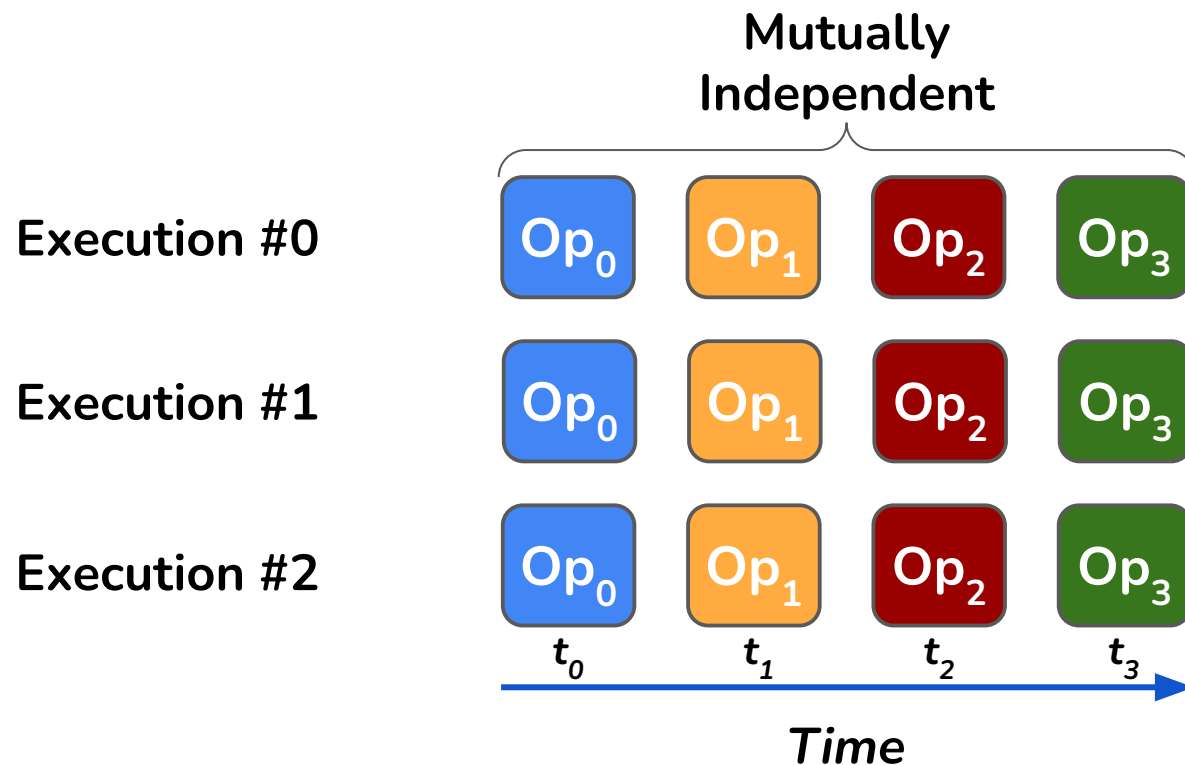
Approximate
Computing
(parameters)



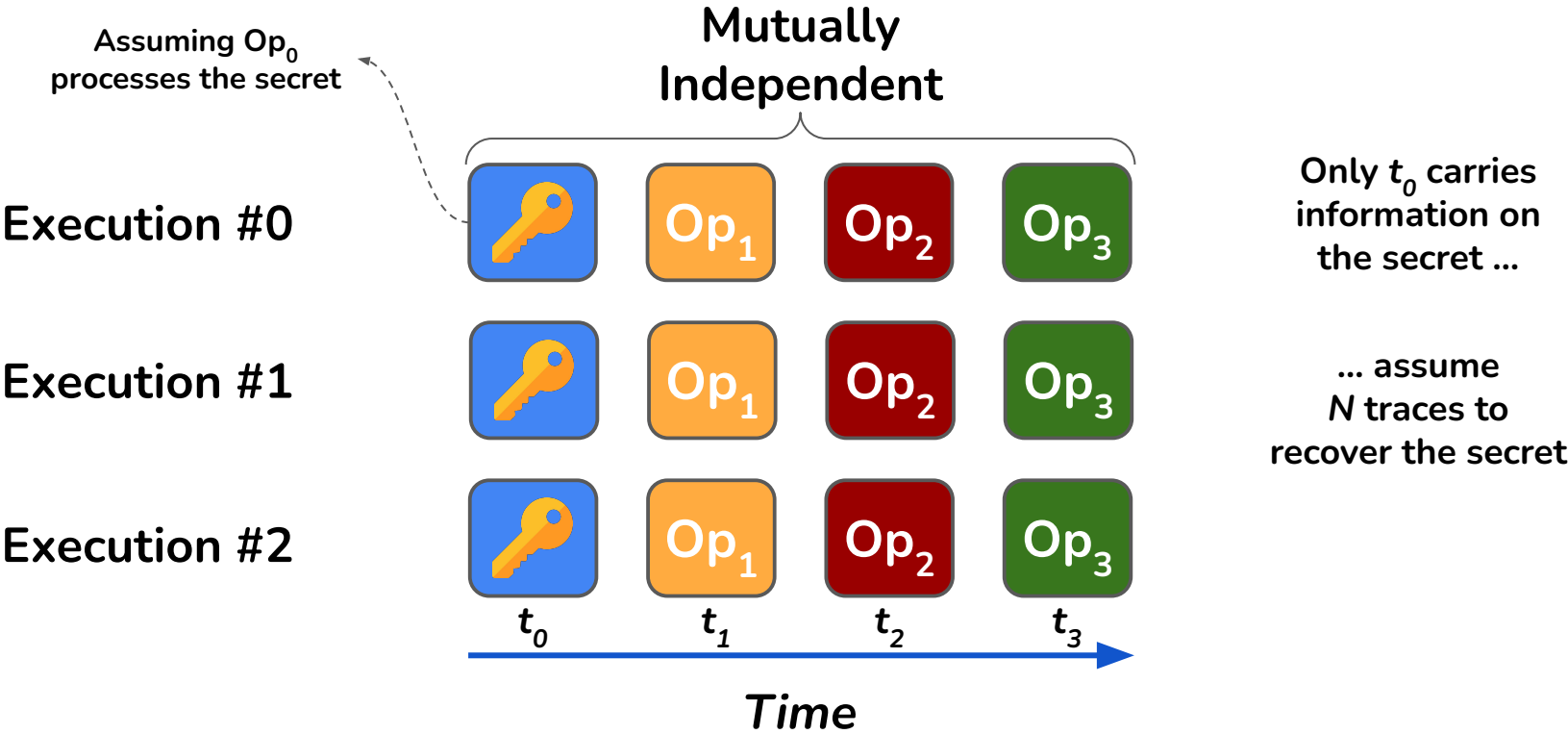
Adversarial Attacks
(hyperparameters)



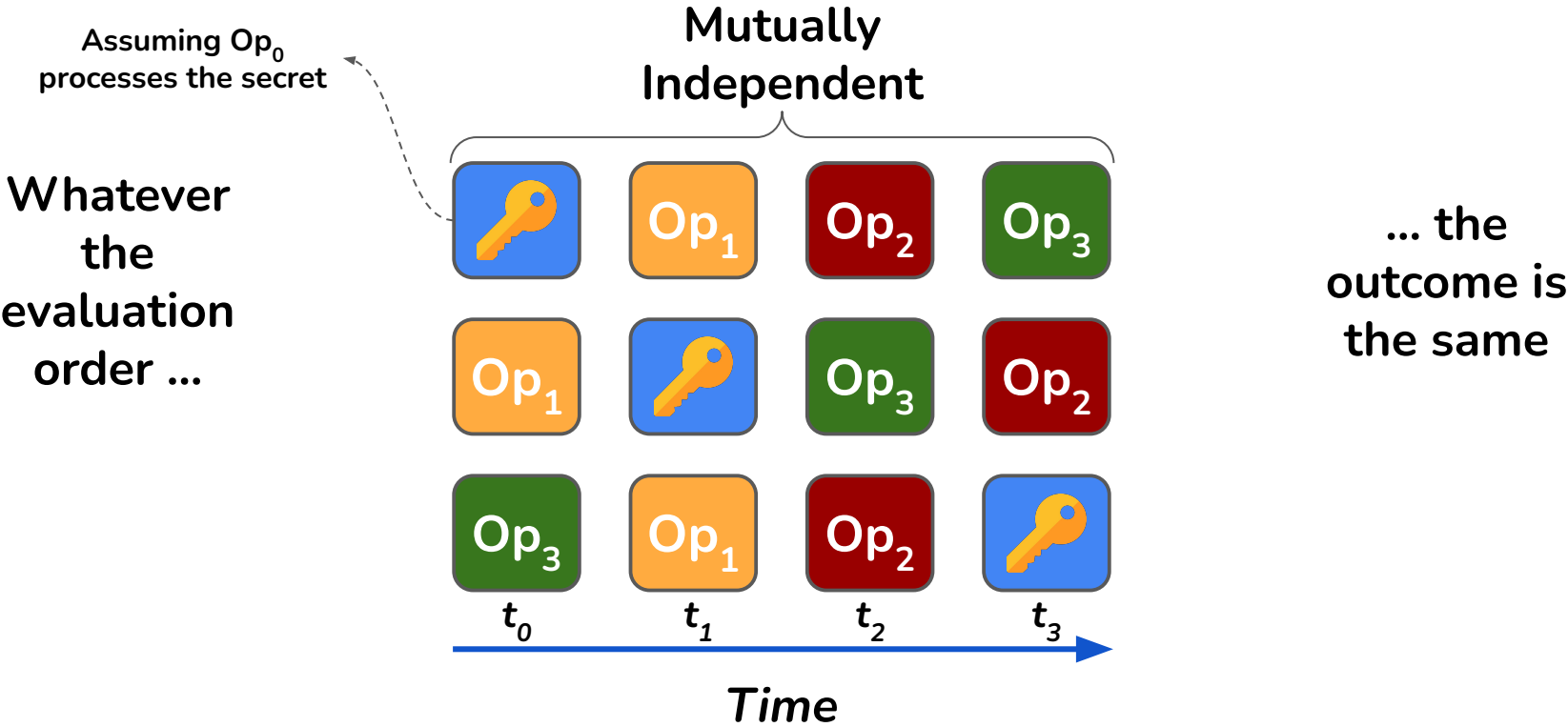
Shuffling – Principle



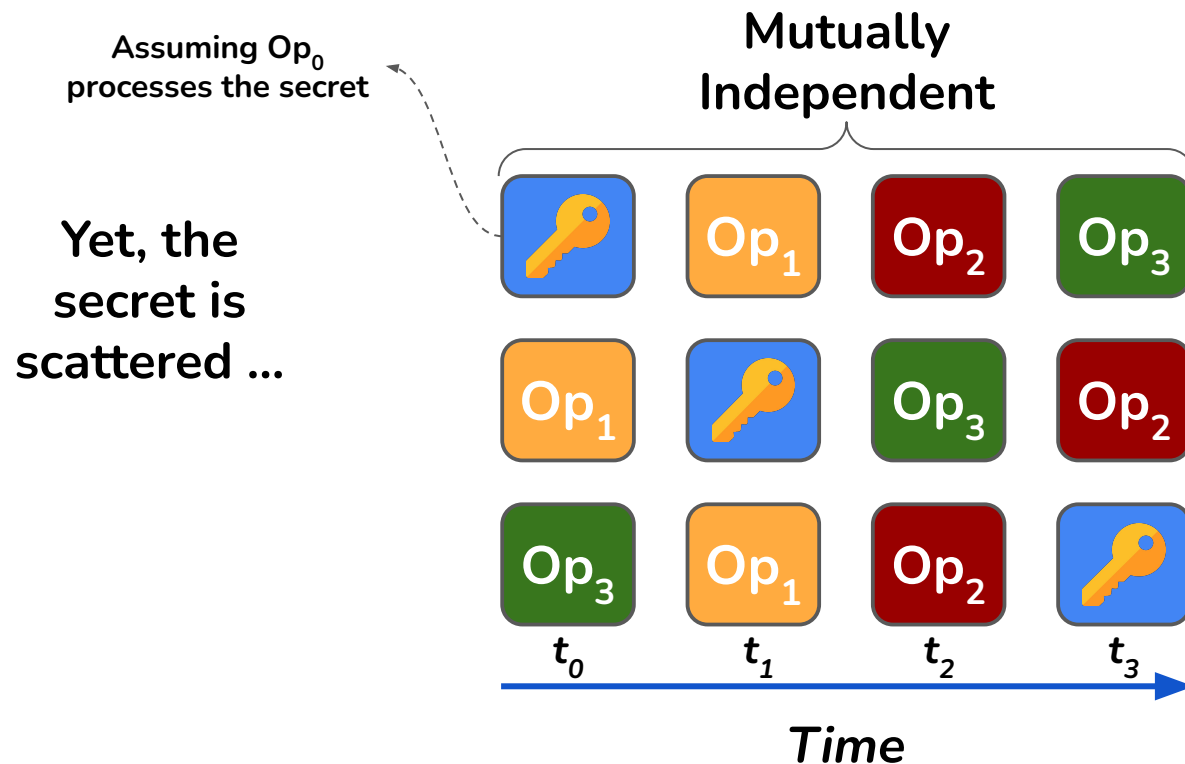
Shuffling – Principle



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Shuffling – Principle

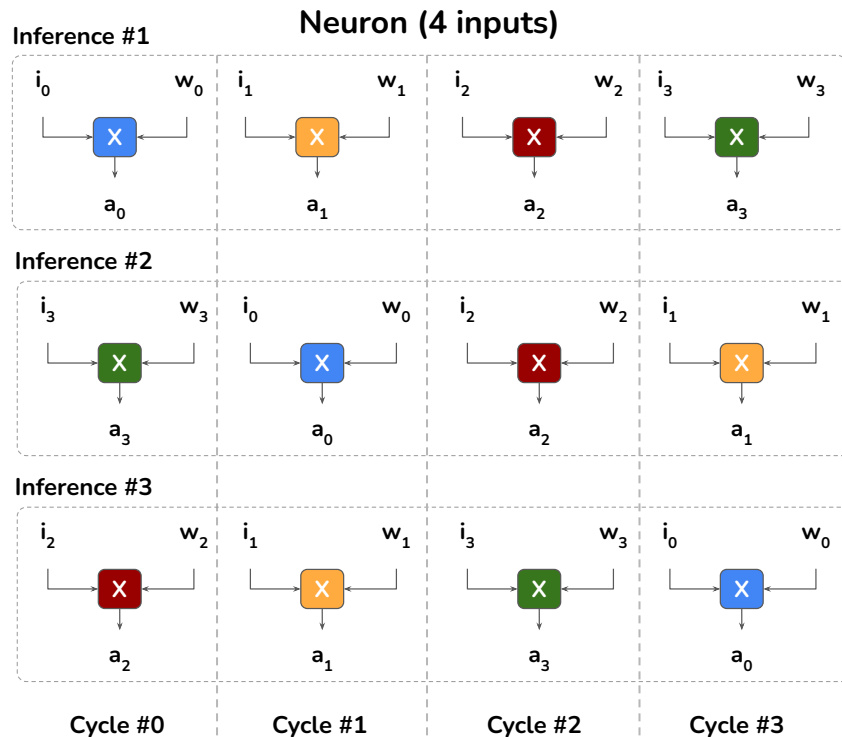


... increasing the attack's complexity [1]:

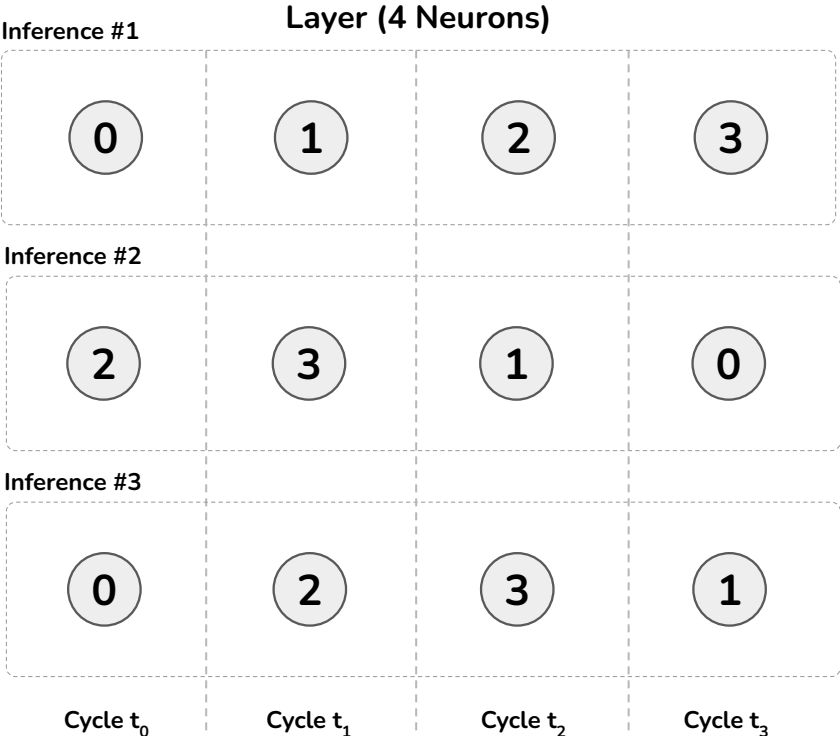
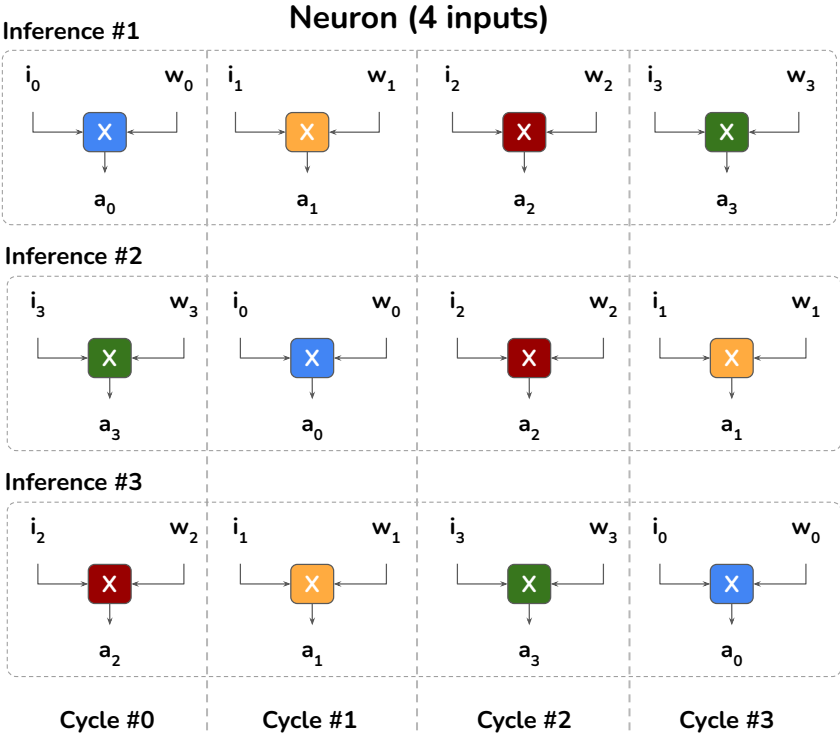
$$\#Traces \propto p^2$$

p : #permutations

Shuffling – Application to DNNs [1]



Shuffling – Application to DNNs [1]



[1] Counteract Side-Channel Analysis of Neural Networks by Shuffling. Brosch et al. ACM DATE 2022.

Shuffling – Advantages and Disadvantages

Advantages:



Data / operations
independent



Any mutually independent
operation



Increased attack
complexity

Disadvantages:



Sufficient number
of permutations



Expensive
design



Permutation
generator
to be protected

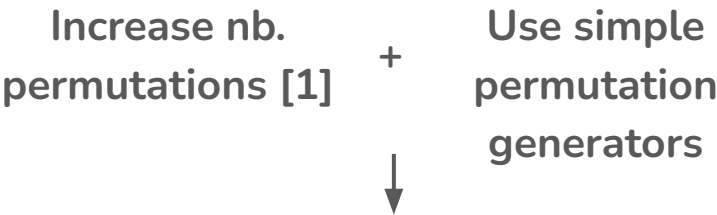


Sensitive to
frequency
analyses



Sensitive to
horizontal
attacks

Shuffling – Research Leads



Find other mutually
independent operations
(e.g., channels shuffling)

Disadvantages:



Sufficient number
of permutations



Expensive
design

[1] Counteract Side-Channel Analysis of Neural Networks by Shuffling. Brosch et al. ACM DATE 2022.

Countermeasure Approaches

Classic Approaches

- Generic
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Masking
(parameters)



Shuffling
(parameters)

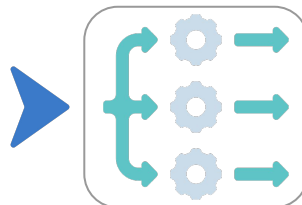


Recent Approaches

Recent Approaches

- DNN-tailored
- First steps

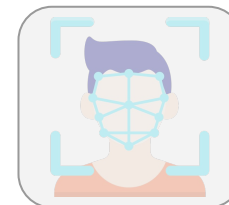
Parallelism
(parameters)



Approximate
Computing
(parameters)

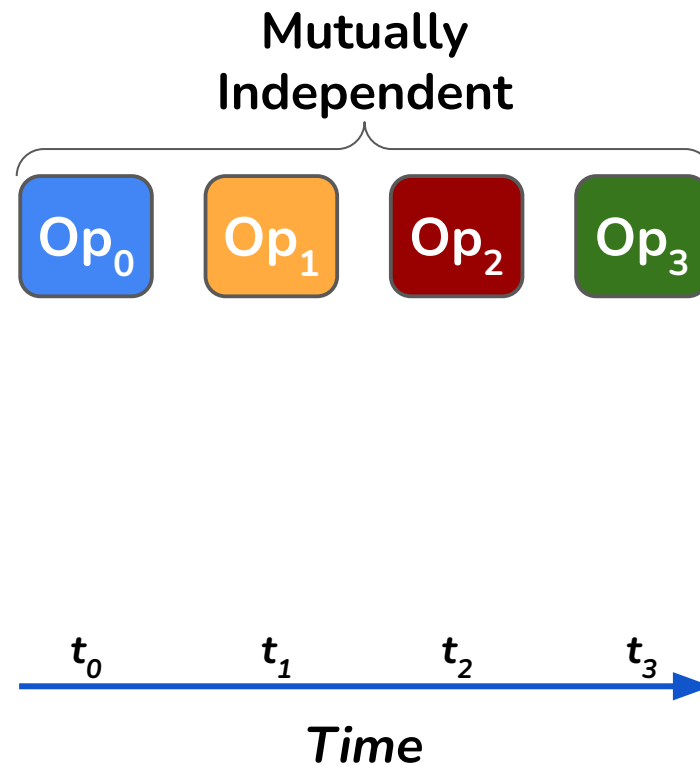


Adversarial Attacks
(hyperparameters)

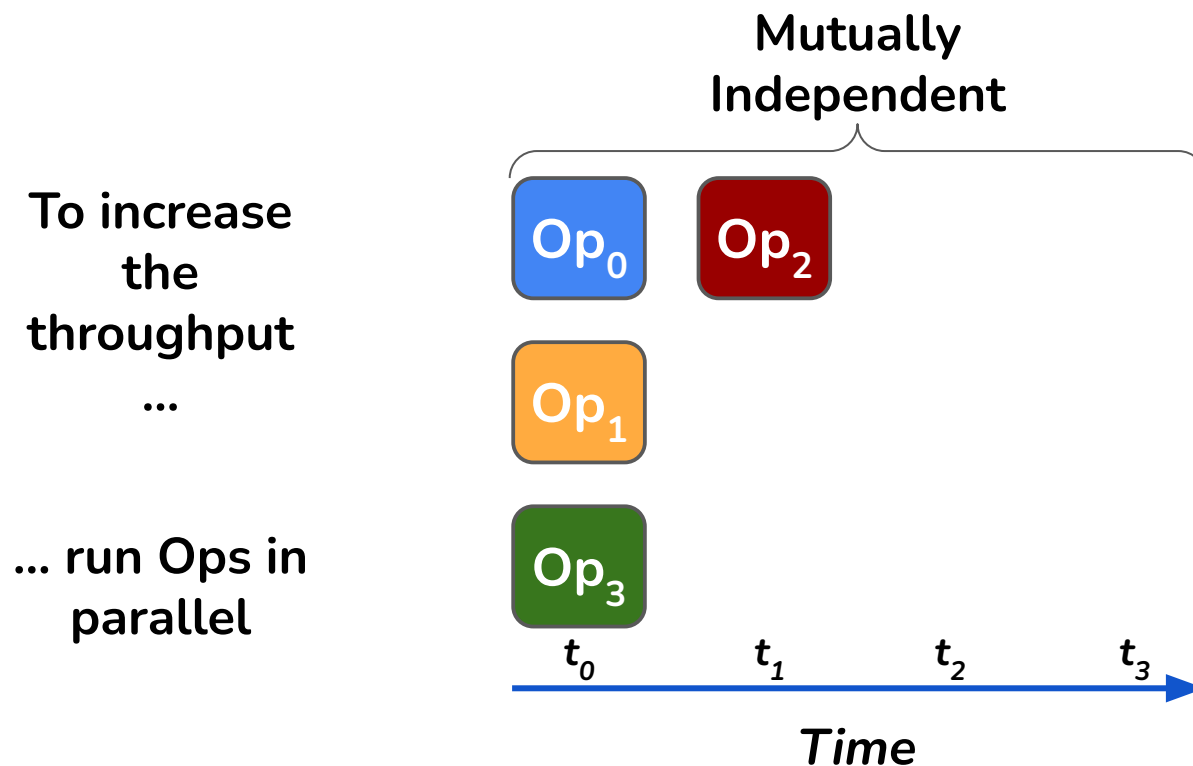


Parallelism – Principle

To increase
the
throughput
...

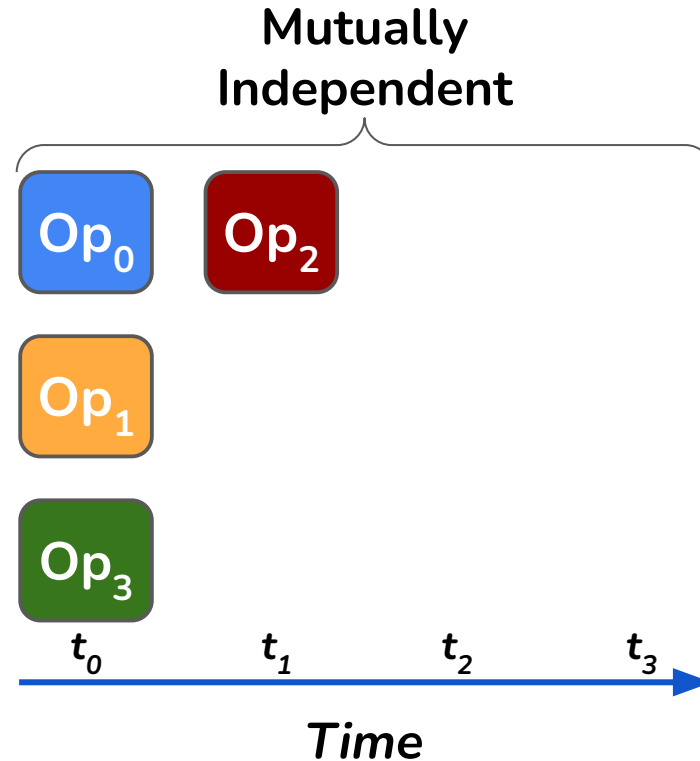


Parallelism – Principle



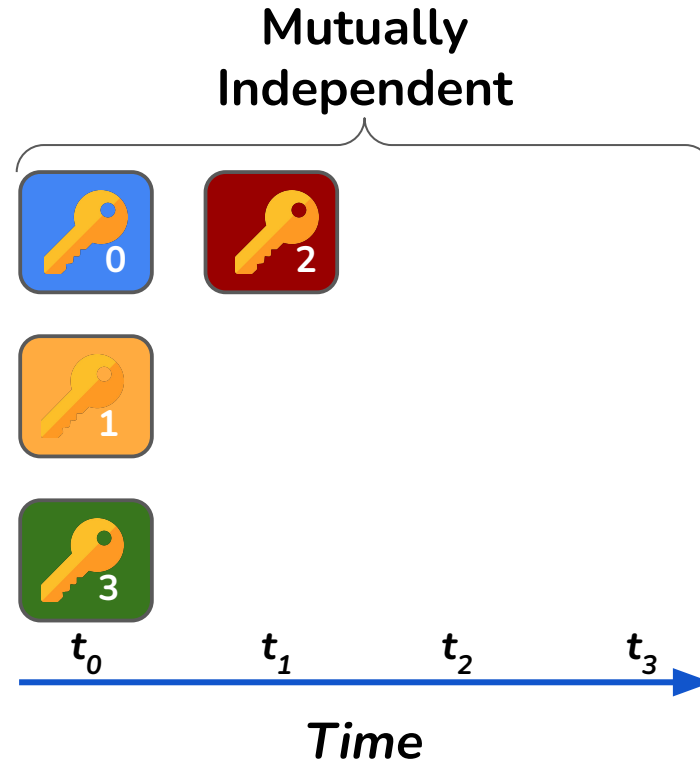
Parallelism – Principle

What
security
implications?



Parallelism – Principle

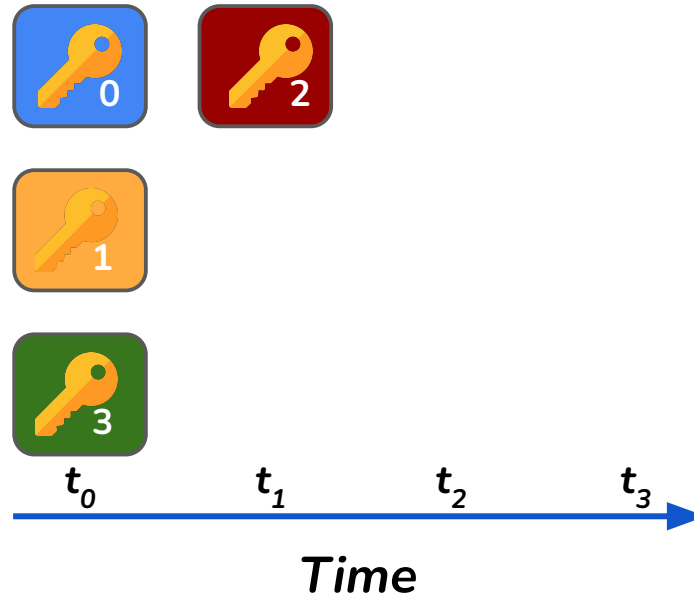
What
security
implications?



Parallelism – Principle

Power consumption at t_0

$$P_{t_0} = P(Op_0) + P(Op_1) + P(Op_3)$$

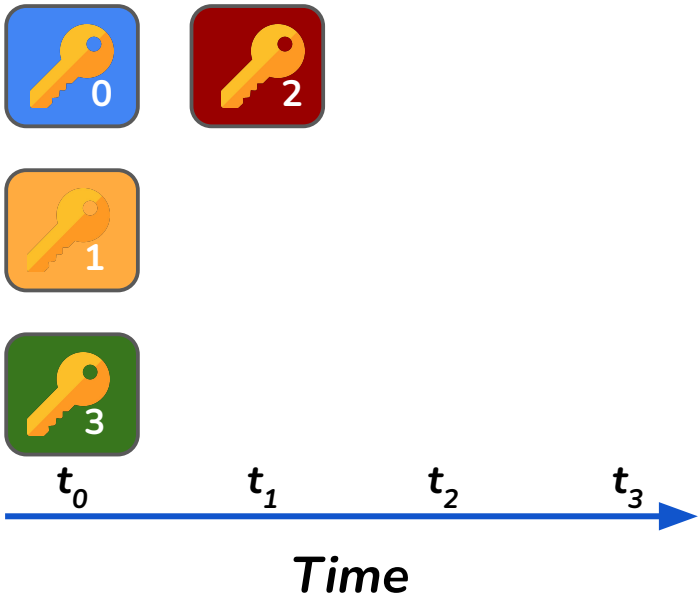


Parallelism – Principle

Power consumption at t_0

$$P_{t_0} = P(f(\text{key}_0, i_0)) + P(f(\text{key}_1, i_1)) + P(f(\text{key}_2, i_2))$$

Ops P mutually independent ...



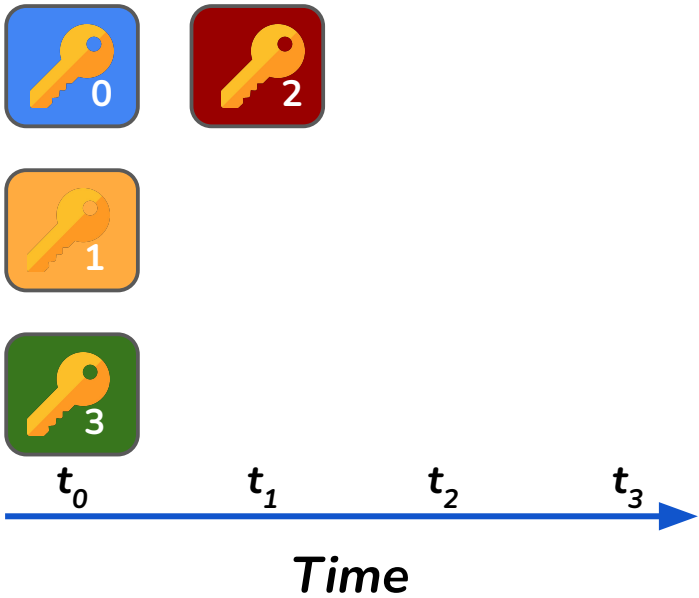
Parallelism – Principle

Power consumption at t_0

$$P_{t_0} = P(f(\text{key}_0, i_0)) + P(f(\text{key}_1, i_1)) + P(f(\text{key}_2, i_2)) = P(f(\text{key}_0, i_0)) + N$$

Ops P mutually independent ...

... attacker can target each secret



Parallelism – Principle

Power consumption at t_0

$$P_{t_0} = P(f(\text{key}_0, i_0)) + P(f(\text{key}_1, i_0)) + P(f(\text{key}_2, i_0)) = \cancel{P(f(\text{key}_0, i_0))} + N$$

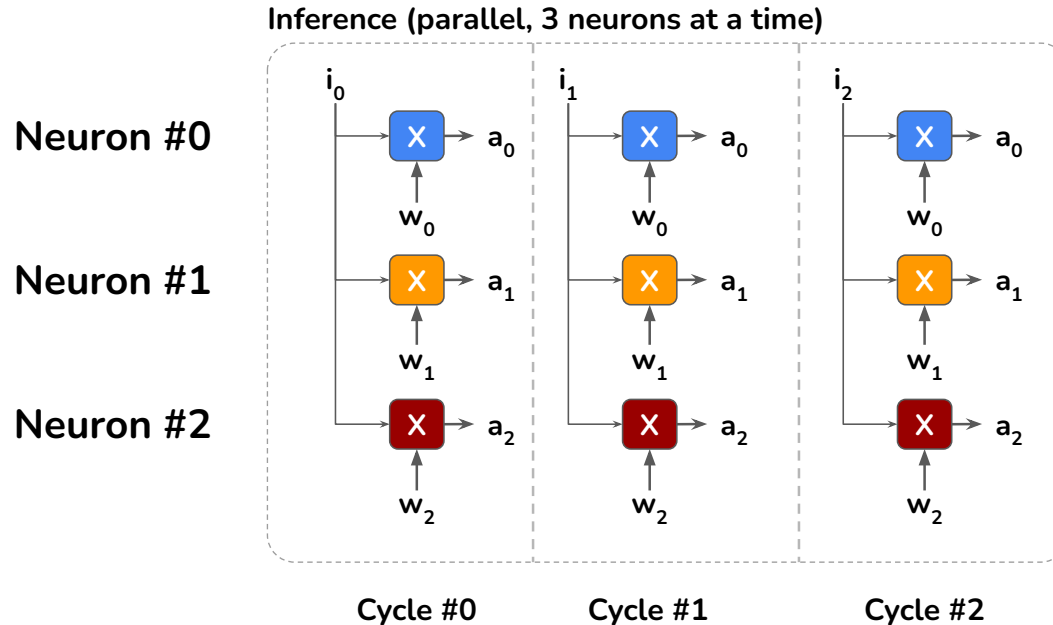
Ops P NOT
mutually
independent ...

... attacker cannot
distinguish
between secrets



Time

Parallelism – Application to DNNs [1]

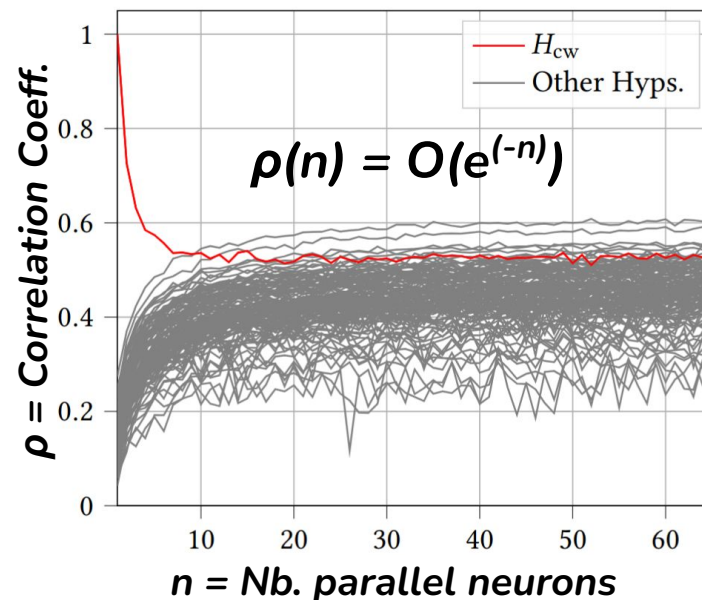


[1] Influence of Parallelism in Vector-Multiplication Units on Correlation Power Analysis. Brosch et al. ACM TECS 2026

Parallelism – Security Evaluation [1]

- ρ : side-channel distinguisher
 - distinguisher: determine the most likely secret's value
- **Red curve**: correct weight value H_{cw}
- **Grey curve**: incorrect weight value
- Higher $n \rightarrow$ difficult to distinguish H_{cw}

Evolution “ ρ vs Nb. parallelism” [1]



Parallelism – Advantages and Disadvantages

Advantages:

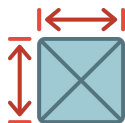


Faster design



Increase attack difficulty

Disadvantages:

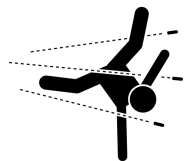


Higher area requirements

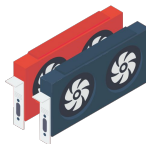


Circuits with limited parallelism (e.g., CPUs)

Parallelism – Research Leads



**How to evade
parallelism?
(e.g., fine-grained
leakage models)**



**May provide a first line
of defence for
GPU-based DNNs?**



**What other operations
may be parallelised?
(e.g., channel-wise
multiplication)**

Countermeasure Approaches

Classic Approaches

- Generic
- Well studied

Masking
(parameters)



Shuffling
(parameters)

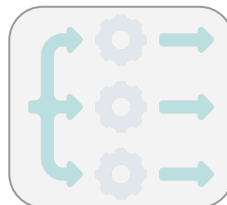


Recent Approaches

Recent Approaches

- DNN-tailored
- First steps

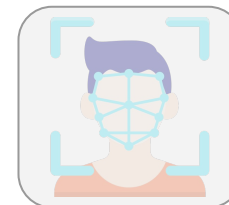
Parallelism
(parameters)



Approximate
Computing
(parameters)



Adversarial Attacks
(hyperparameters)



Approximate Computing (AxC)

Definition [1]

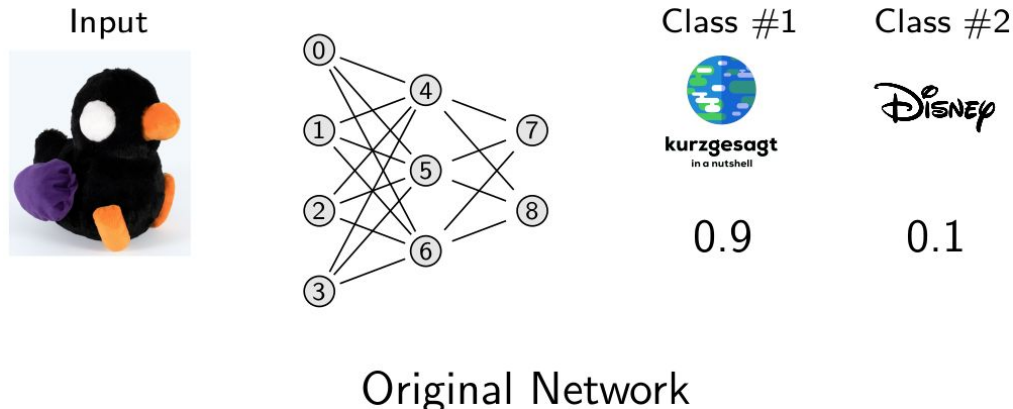
Intentional insertion of errors to trade tunable accuracy loss for valuable resource gains.

Approximate Computing (AxC)

Definition [1]

Intentional insertion of errors to trade tunable accuracy loss for valuable resource gains.

Pruning



Approximate Computing (AxC)

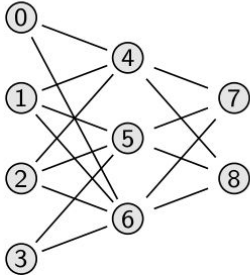
Definition [1]

Intentional insertion of errors to trade tunable accuracy loss for valuable resource gains.

Pruning

Input





Class #1



~~0.9~~
0.7

Class #2



~~0.1~~
0.3

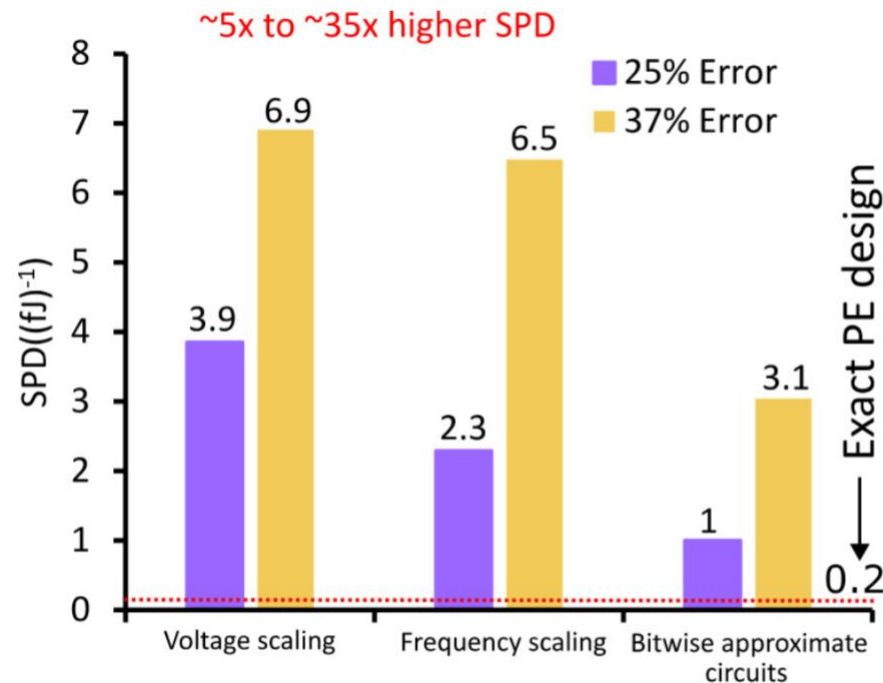
Pruned Network

[1] Approximate Computing Survey, Part I: Terminology and Software & Hardware Approximation Techniques. Vasileios L. et al. ACM Computing Surveys. 2025.

AxC – Leakage Interaction

- **Voltage scaling [1]:**
Reduce supplied voltage
 - Reduce power consumption
- **Frequency scaling [1]:**
Increase clock frequency
 - Reduce latency
- **Approximated multipliers [1, 2]:**
 - Reduce latency
 - Reduce power consumption

$$SPD = \frac{\#Traces}{Power \cdot Latency}$$



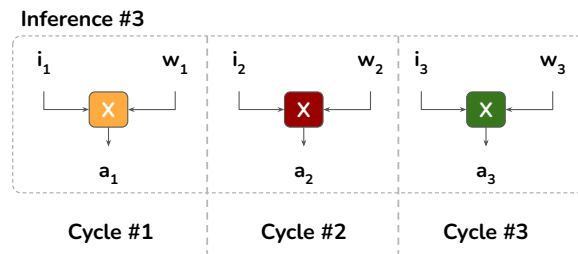
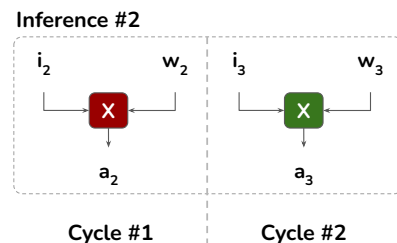
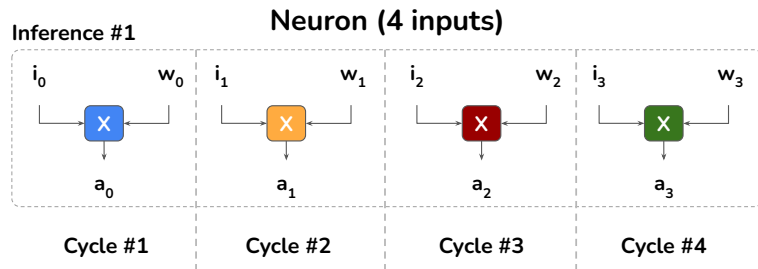
Security-Power-Delay (SPD) of three AxC techniques, for different precision error rates [1]

[1] Security of Approximate Neural Networks against Power Side-channel Attacks

[2] ApproPower: Characterizing the Power Side-Channel Features of Approximate Multipliers with Symbolic Path Analysis

AxC – Countermeasure Design

MACPruning [1]



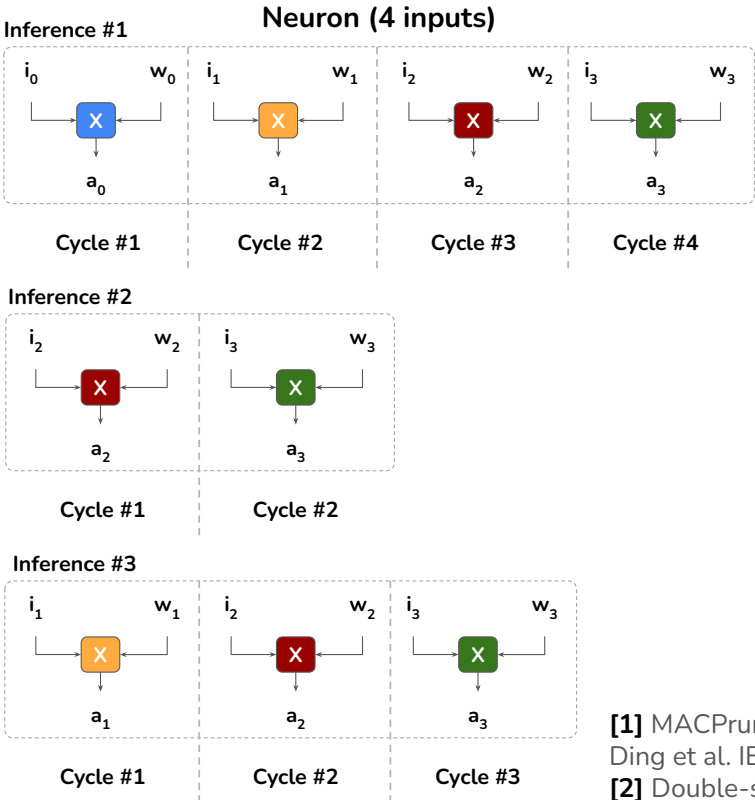
Randomly skip multiplications / weights

- w_i recovered with $2^{(i+1)}N$ traces
- Mean inference latency:

$$\mu_{\text{Latency}} = \frac{1}{2^L} \sum_{i=0}^L \binom{L}{i} (L-i)$$

Latency of unprotected DNN

AxC – Countermeasure Design



MACPruning [1]

Randomly skip multiplications / weights

- w_i recovered with $2^{(i+1)}N$ traces
- Mean inference latency:

$$\mu_{\text{Latency}} = \frac{1}{2^L} \sum_{i=0}^L \binom{L}{i} (L-i)$$

Latency of unprotected DNN

MACPruning Circumvention [2]

MACPruning hints what weights are skipped

[1] MACPruning: Dynamic Operation Pruning to Mitigate Side-Channel DNN Model Extraction. Ding et al. IEEE HOST 2025.

[2] Double-strike: Breaking Approximation-based Side-channel Countermeasures for DNNs. Casalino et al. IEEE HOST 2026.

AxC – Advantages and Disadvantages

Advantages:



Light countermeasure



Increase attack difficulty

Disadvantages:



Not suitable in
certain cases
(e.g., voltage scaling)



May unintentionally leak
information

AxC – Research Leads

What AxC techniques can
underpin a countermeasure?



What properties to be
satisfied?

Disadvantages:



Not suitable in
certain cases
(e.g., voltage scaling)



May unintentionally leak
information

Countermeasure Approaches

Classic Approaches

- Generic
- Well studied

Masking
(parameters)



Shuffling
(parameters)



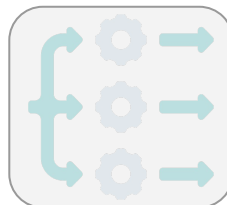
Classic Approaches

Recent Approaches

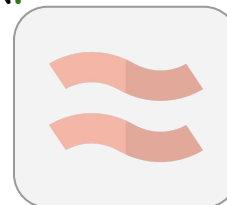
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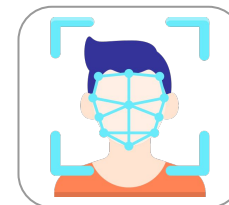
Parallelism
(parameters)



Approximate
Computing
(parameters)

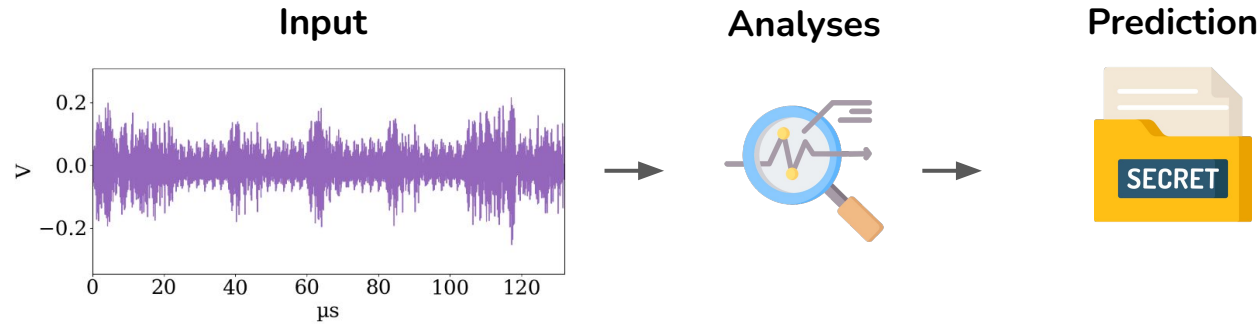


Adversarial Attacks
(hyperparameters)



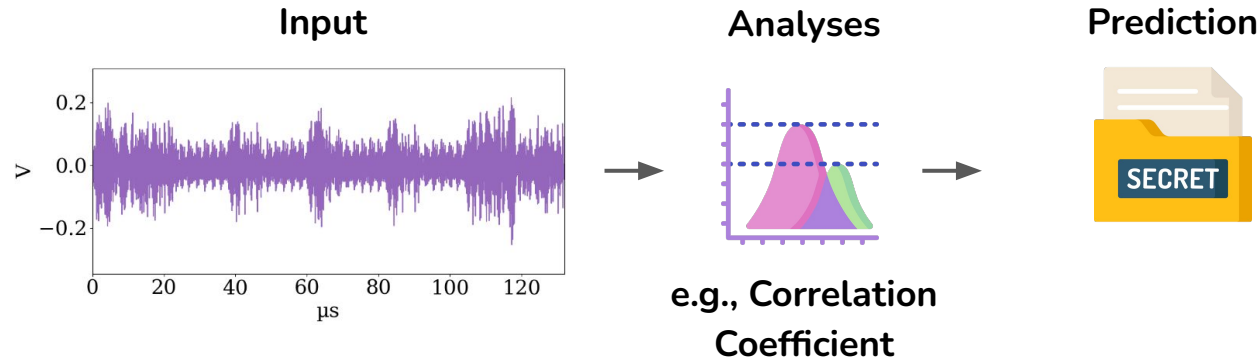
Adversarial Attacks

- Side-channel analysis as a classification problem



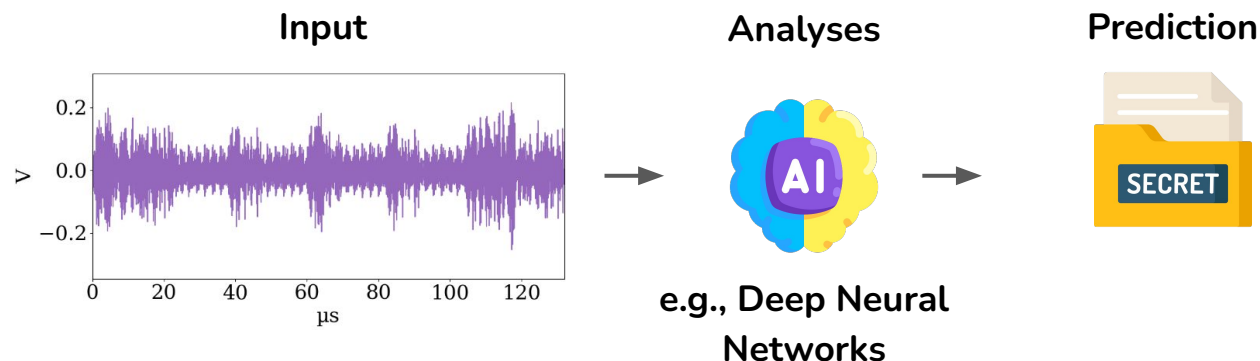
Adversarial Attacks

- Side-channel analysis as a classification problem



Adversarial Attacks

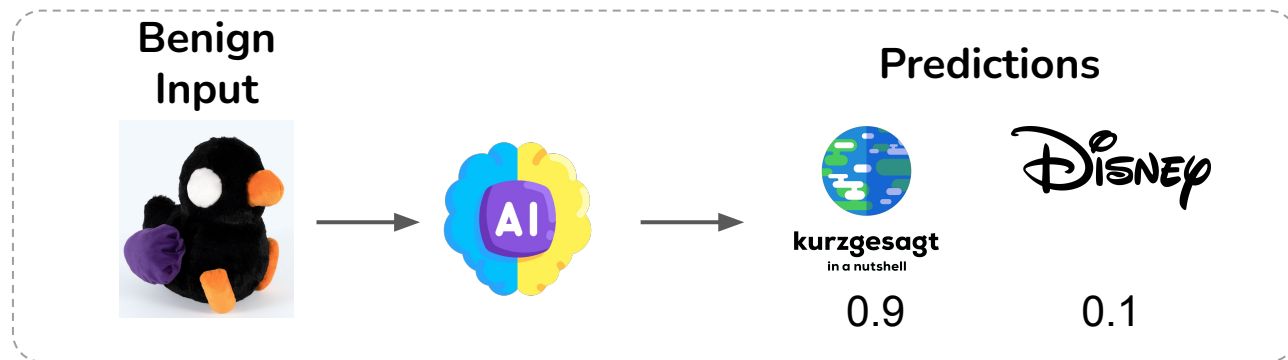
- Side-channel analysis as a classification problem



- Deep neural networks susceptible to *adversarial attacks*
- Can we use such attacks as a countermeasure?

Adversarial Attacks

Correct Classification



Adversarial Attacks

Correct Classification

Benign
Input



Predictions



kurzgesagt
in a nutshell

0.9

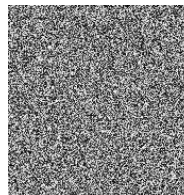
Disney

0.1

+

Misclassification

Imperceptible
noise



=

Adversarial
Input



Predictions



kurzgesagt
in a nutshell

0.3

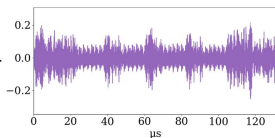
Disney

0.7

Adversarial Attacks

Correct Classification

Benign
Input



Predictions



0.9

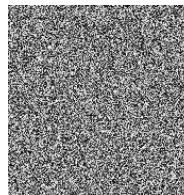


0.1

+

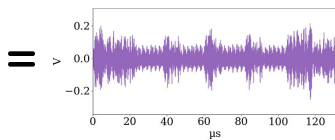
Misclassification

Imperceptible
noise



=

Adversarial
Input



Predictions

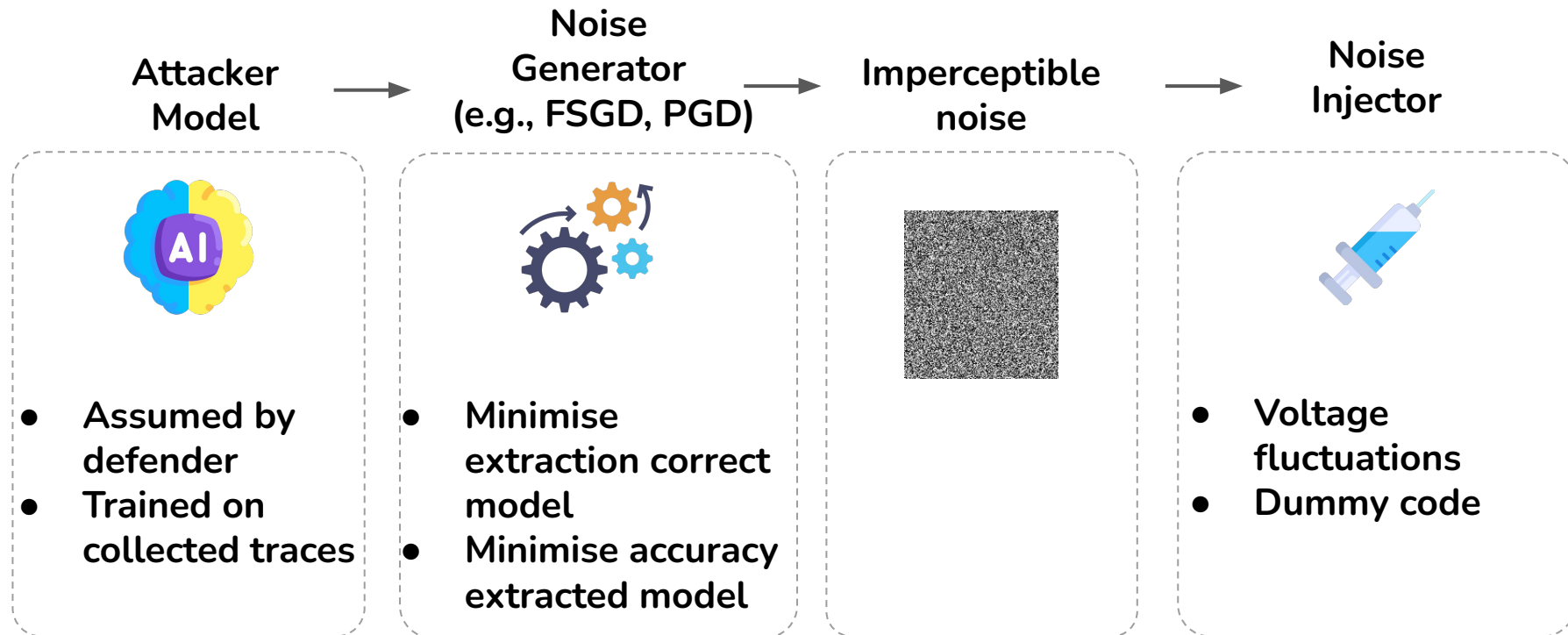


0.3



0.7

Adversarial Attacks [1]



[1] AdvProtego: A Unified Framework to SafeGuard Deep Learning Models Against Side Channel Leakage. Xiaobei et al. IEEE HOST 2026

Adversarial Attacks – Advantages and Disadvantages

Advantages:



Defeat ML-based
attackers



Light countermeasure

Disadvantages:



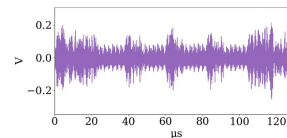
Update victim model



Update countermeasure



Dependence
on attacker
model



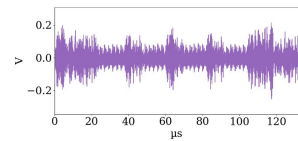
Dependence
on collected
traces



Dependence
on noise
generation

Adversarial Attacks – Research Leads

Reduce dependency



Acquire traces
with different
setups

Transfer
learning

Transferable
perturbation



Does it thwart simpler
attacks?
(e.g., correlation attacks)

Countermeasure Evasion



Can approximated DNNs
circumvent this
countermeasure? [1]



Protecting parameters

[1] Defensive approximation: securing CNNs using approximate computing.
Guesmi A. ACM ASPLOS 2021.

Some comparisons ...

Masking
(parameters)



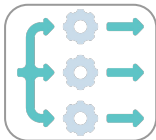
Provable secure
Very expensive design

Shuffling
(parameters)



Increase security
Expensive design

Parallelism
(parameters)



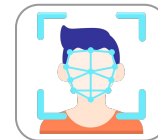
Fast and secure
Expensive design
Specific attacker model

AxC
(parameters)



Lightweight and secure
Specific HW support
May leave traces

Adversarial Attacks
(hyperparameters)



Thwart ML-based attacks
Lightweight
Huge dependencies
Specific attacker model

... and some questions

**Masking
(parameters)**



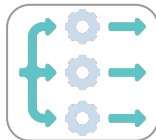
How to reduce its cost?

**Shuffling
(parameters)**



What other operations?

**Parallelism
(parameters)**



**How to evade such
parallelism?
What about ML-based
attackers?**

**AxC
(parameters)**



**What security
guarantees?**

**Adversarial Attacks
(hyperparameters)**



**What about parameters?
What about simpler
attacks?**

Unity is Strength

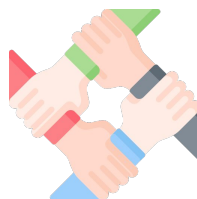
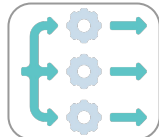
Masking



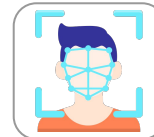
Shuffling



Parallelism



Adversarial Attacks



AxC





Thank you!



Any Questions?

`lorenzo.casalino@{centralesupelec, irisa}.fr`